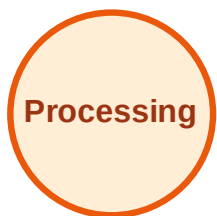


BFS Traversal

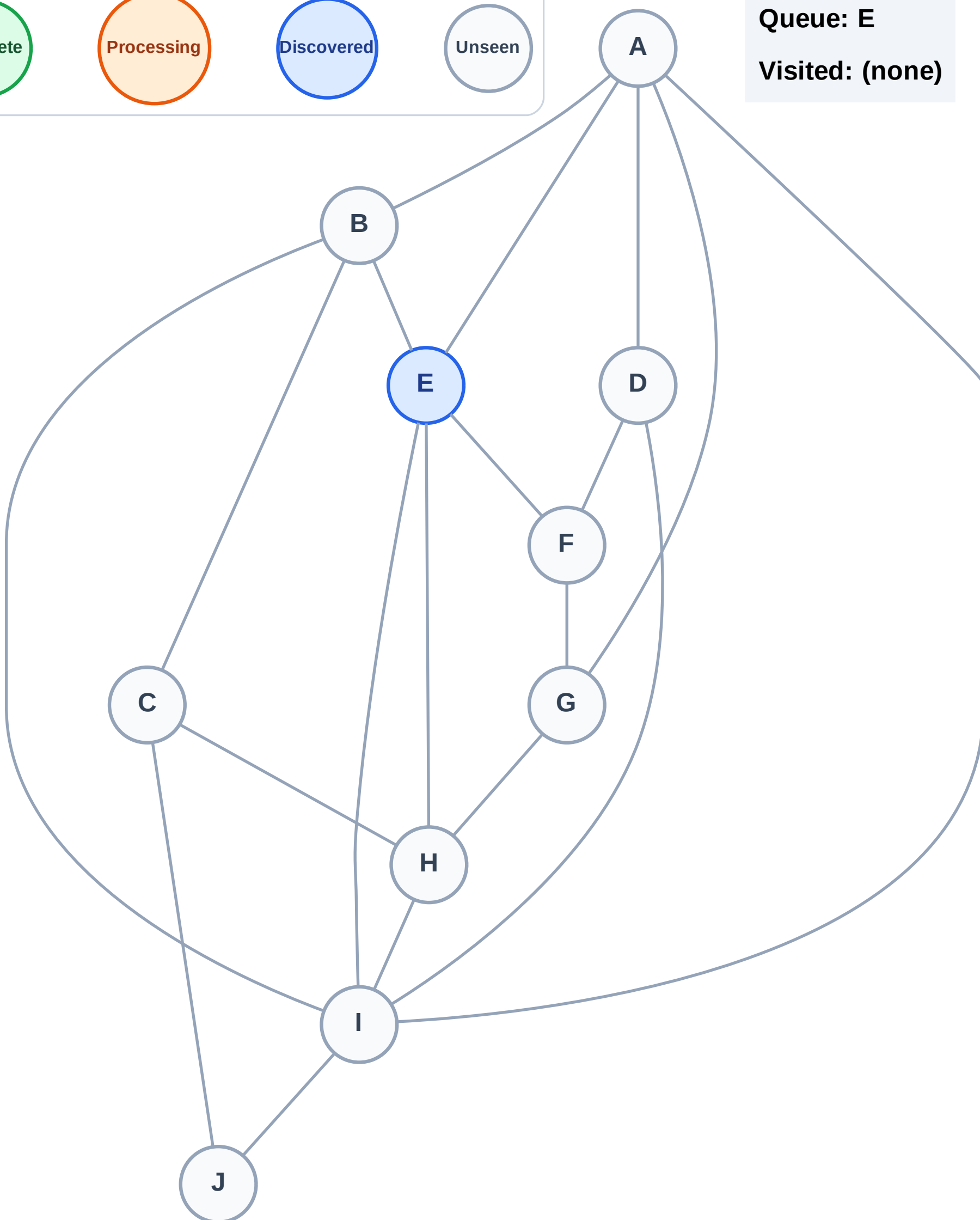
Step 1: Start at E

Legend



Queue: E

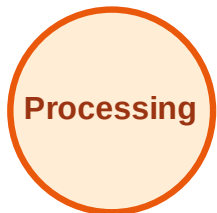
Visited: (none)



BFS Traversal

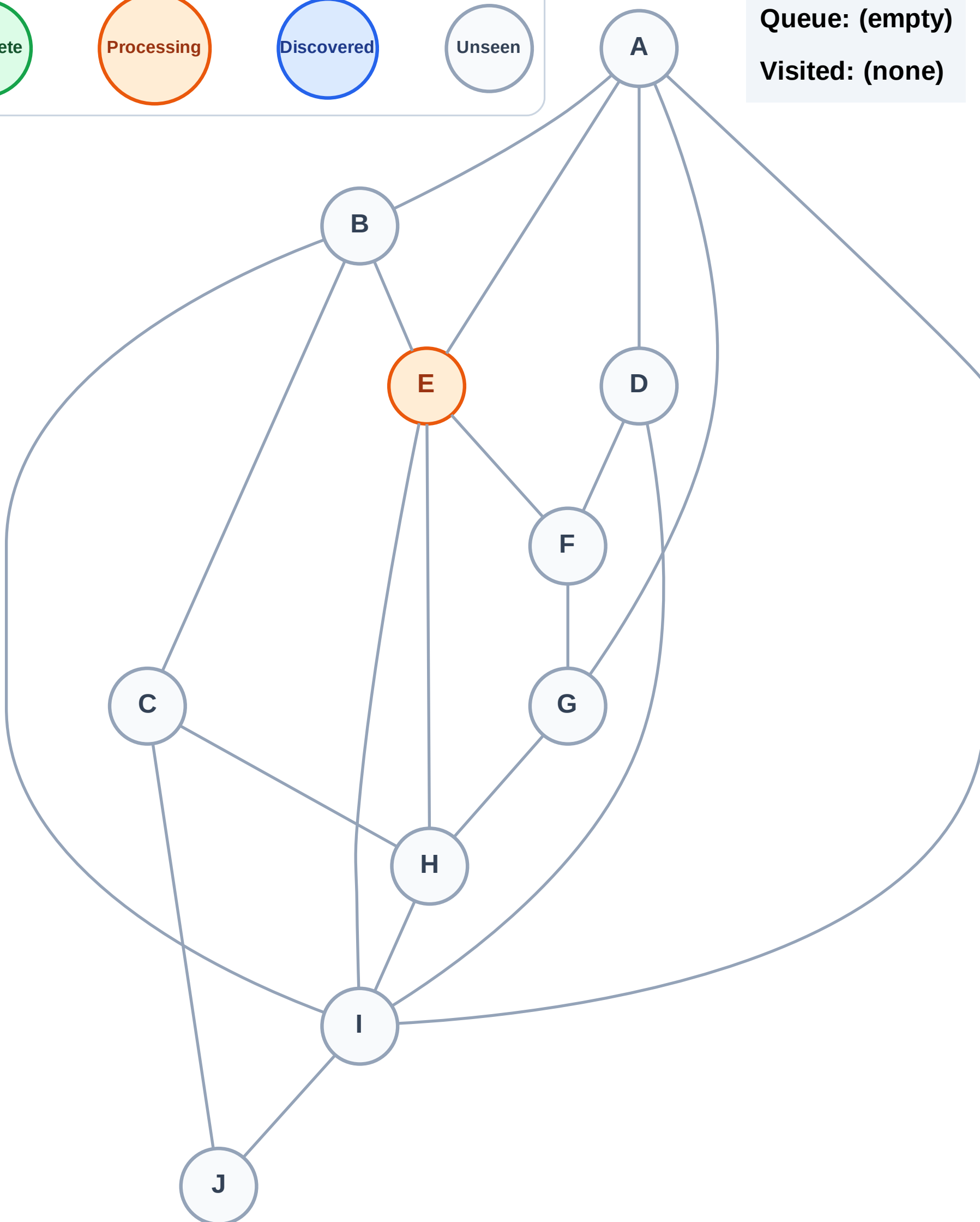
Step 2: Process node E

Legend



Queue: (empty)

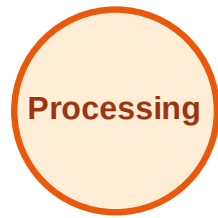
Visited: (none)



BFS Traversal

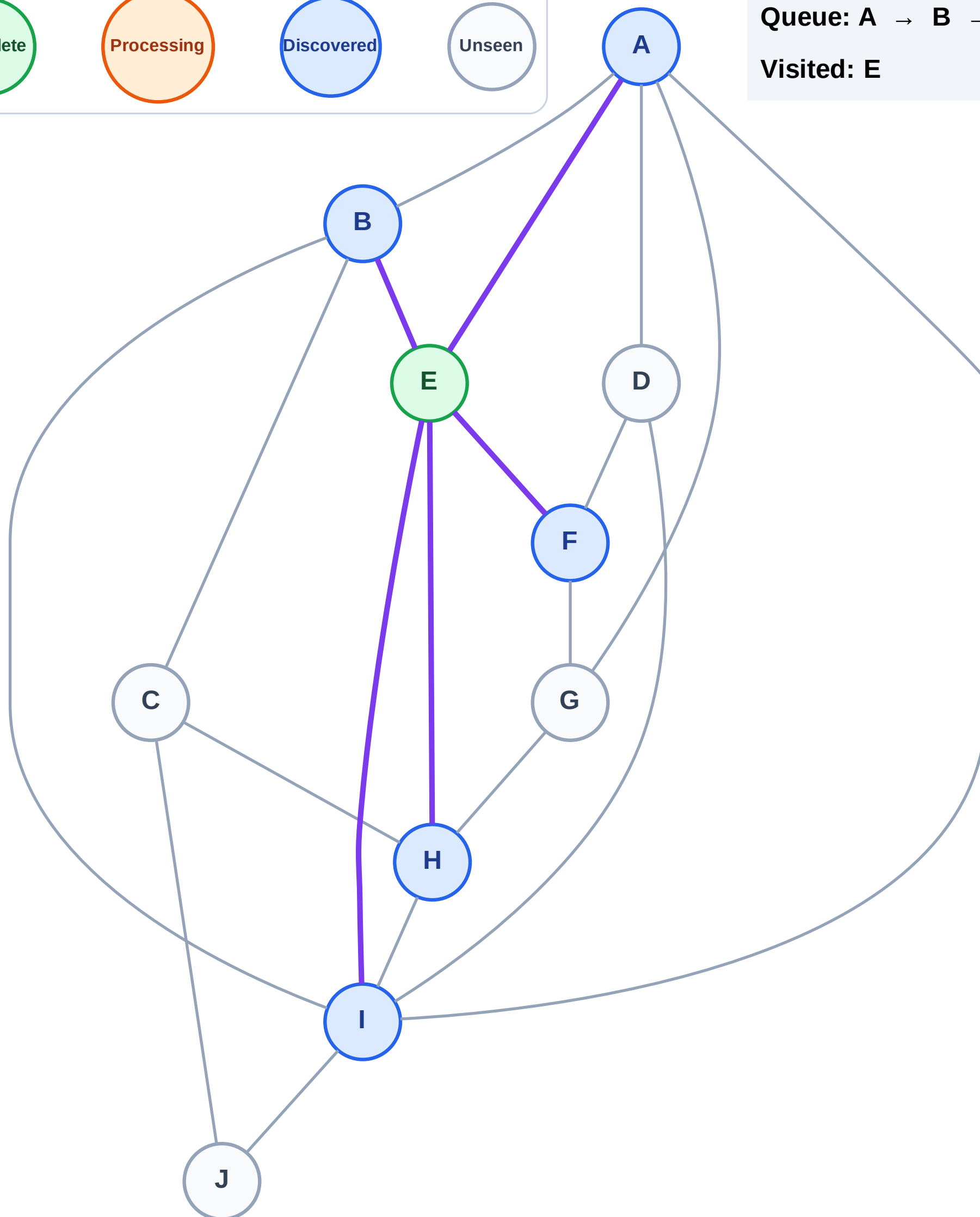
Step 3: Discover A, B, F, H, I from E

Legend



Queue: A → B → F → H → I

Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

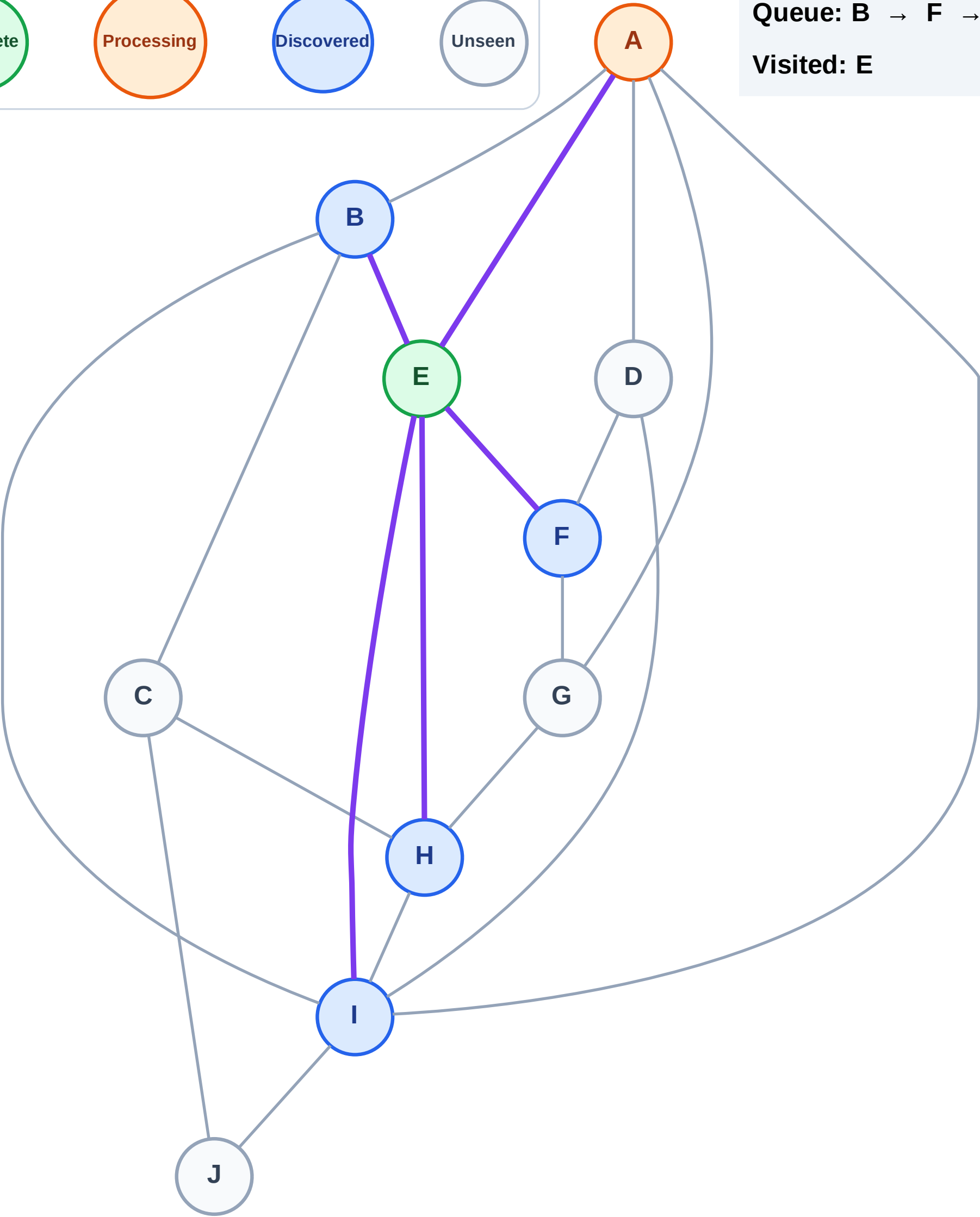
Processing

Discovered

Unseen

Queue: B → F → H → I

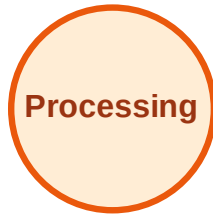
Visited: E



BFS Traversal

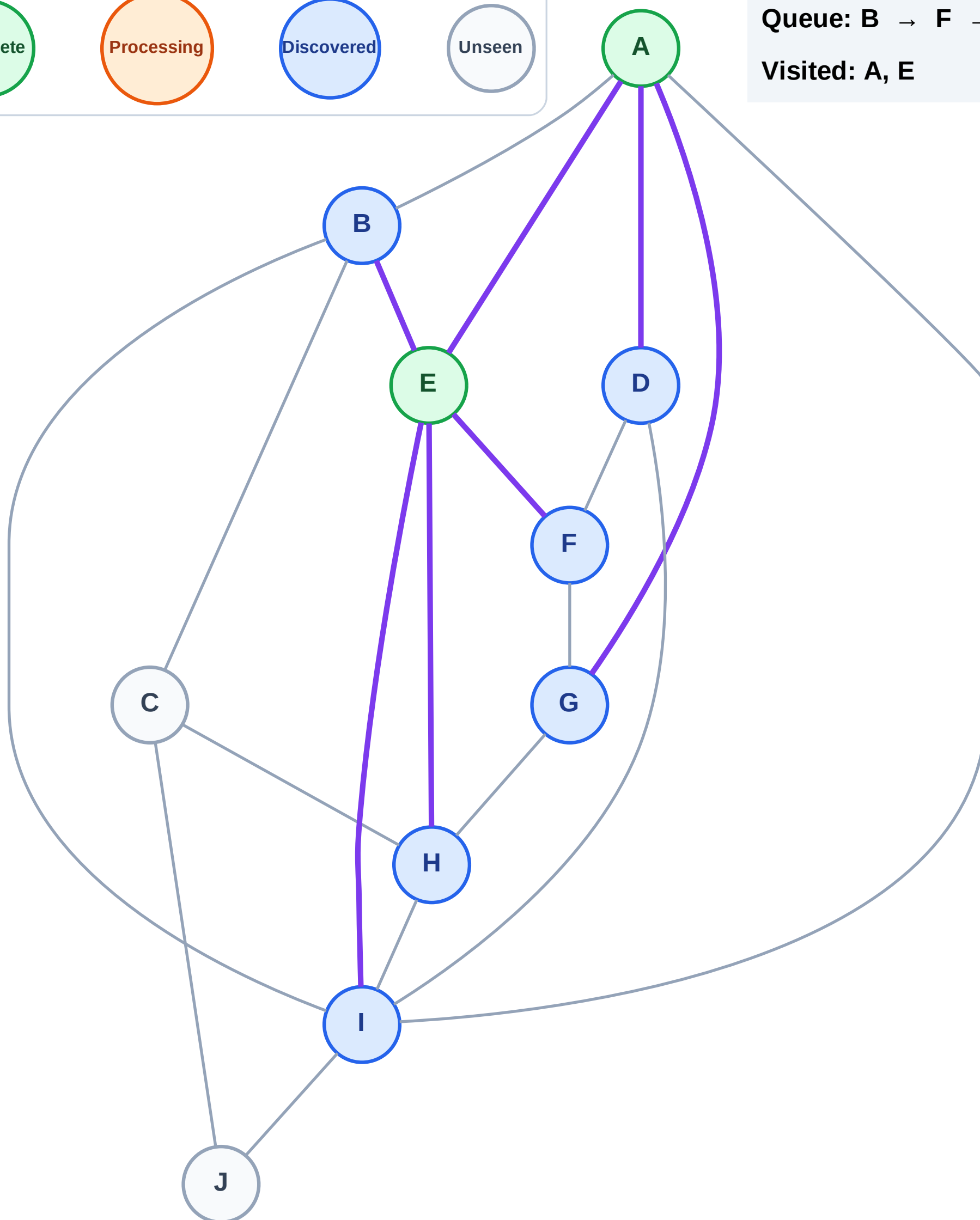
Step 5: Discover D, G from A

Legend



Queue: B → F → H → I → D → G

Visited: A, E



BFS Traversal

Step 6: Process node B

Legend

Complete

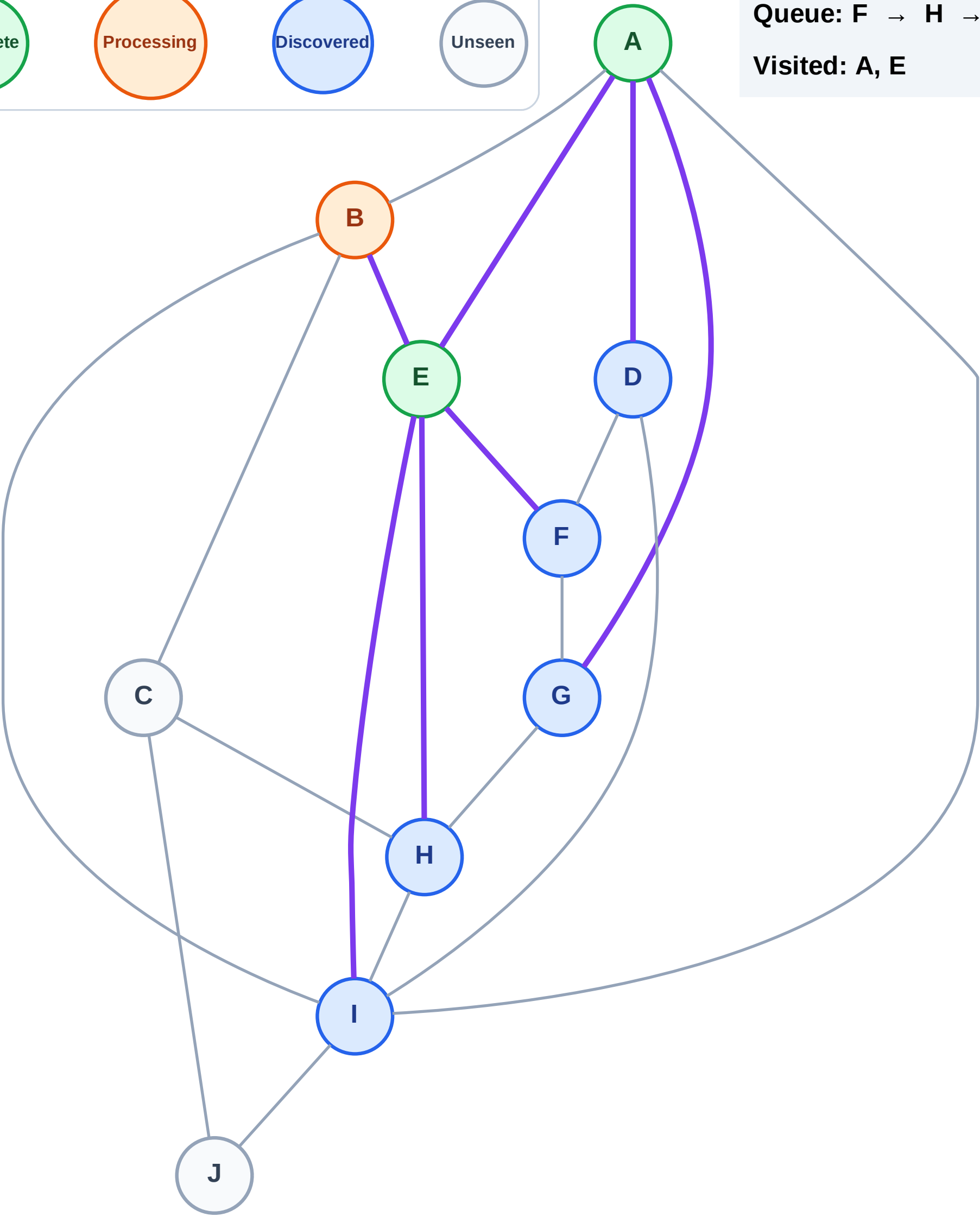
Processing

Discovered

Unseen

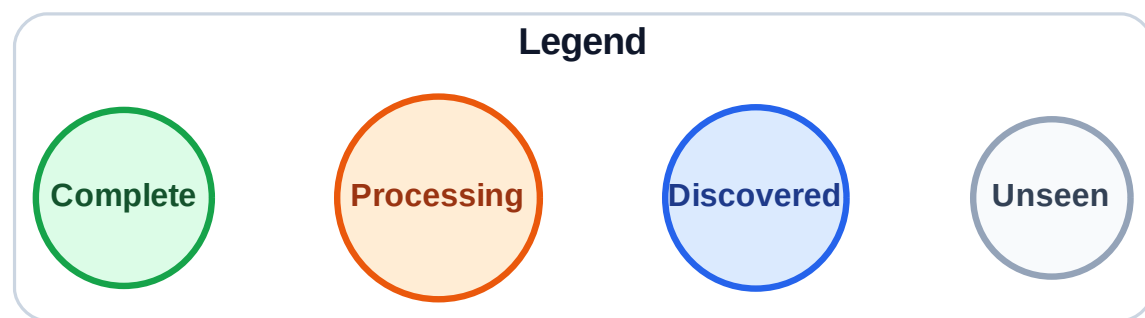
Queue: F → H → I → D → G

Visited: A, E



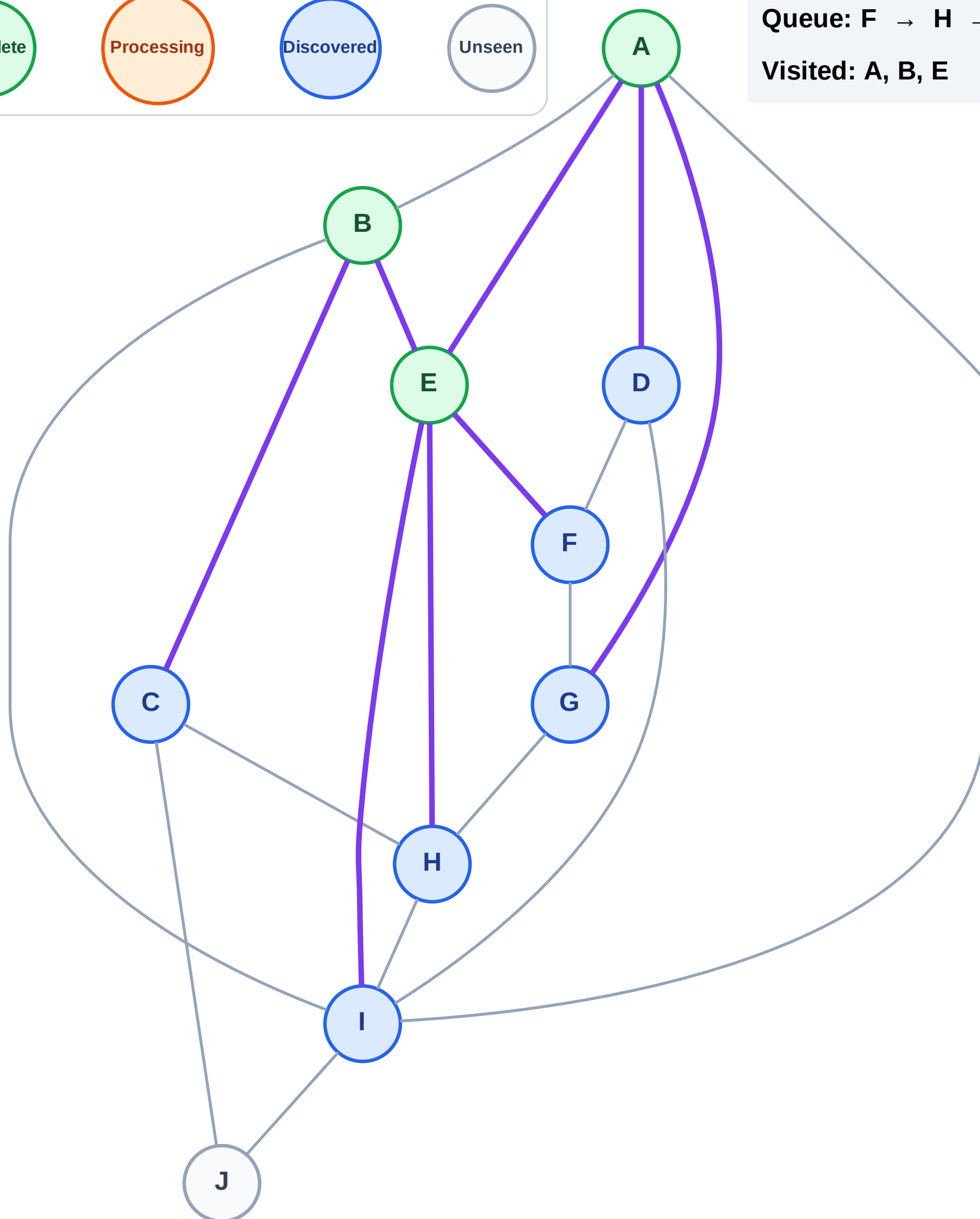
BFS Traversal

Step 7: Discover C from B



Queue: F → H → I → D → G → C

Visited: A, B, E



BFS Traversal

Step 8: Process node F

Legend

Complete

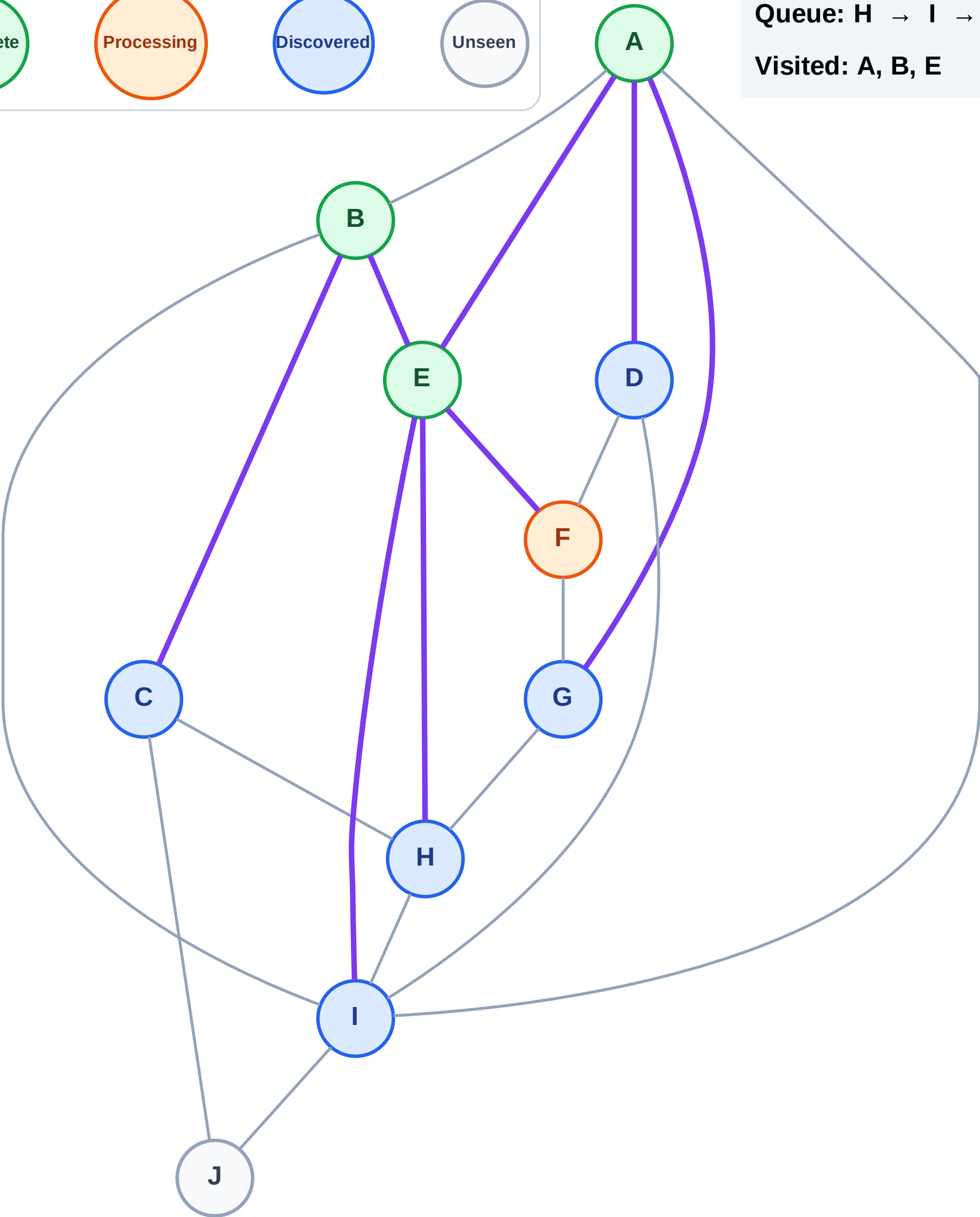
Processing

Discovered

Unseen

Queue: H → I → D → G → C

Visited: A, B, E



BFS Traversal

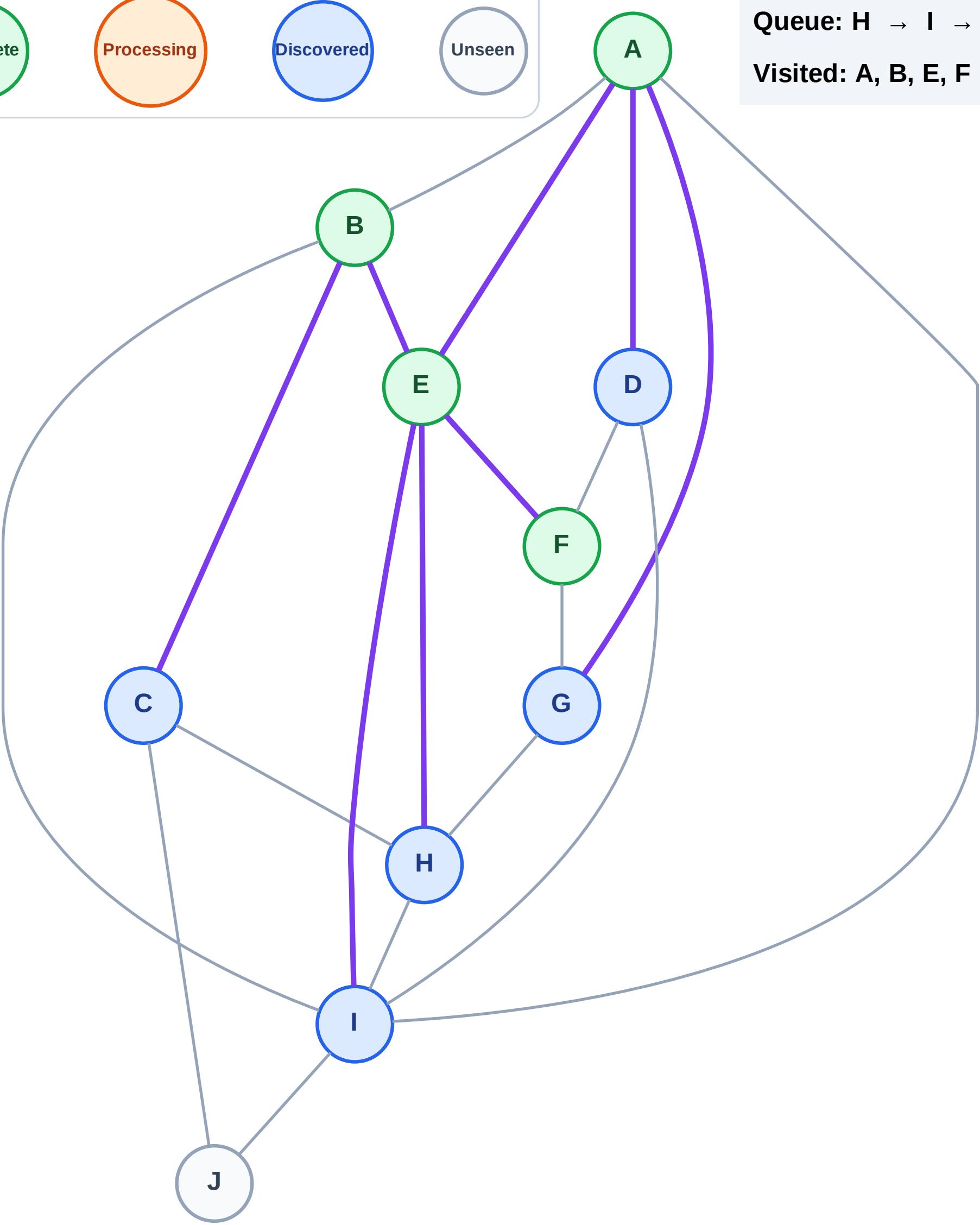
Step 9: No new neighbors from F

Legend



Queue: H → I → D → G → C

Visited: A, B, E, F



BFS Traversal

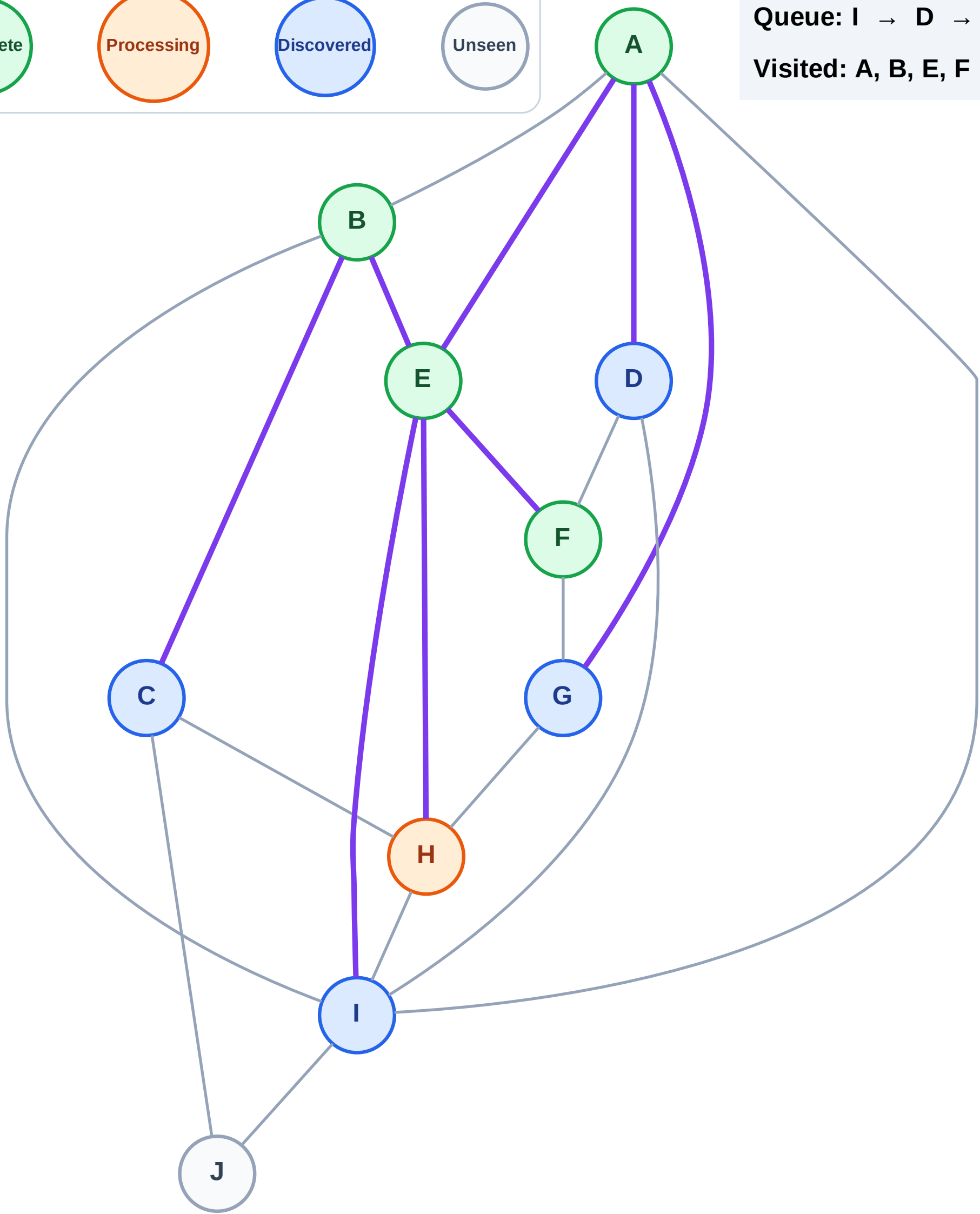
Step 10: Process node H

Legend



Queue: I → D → G → C

Visited: A, B, E, F



BFS Traversal

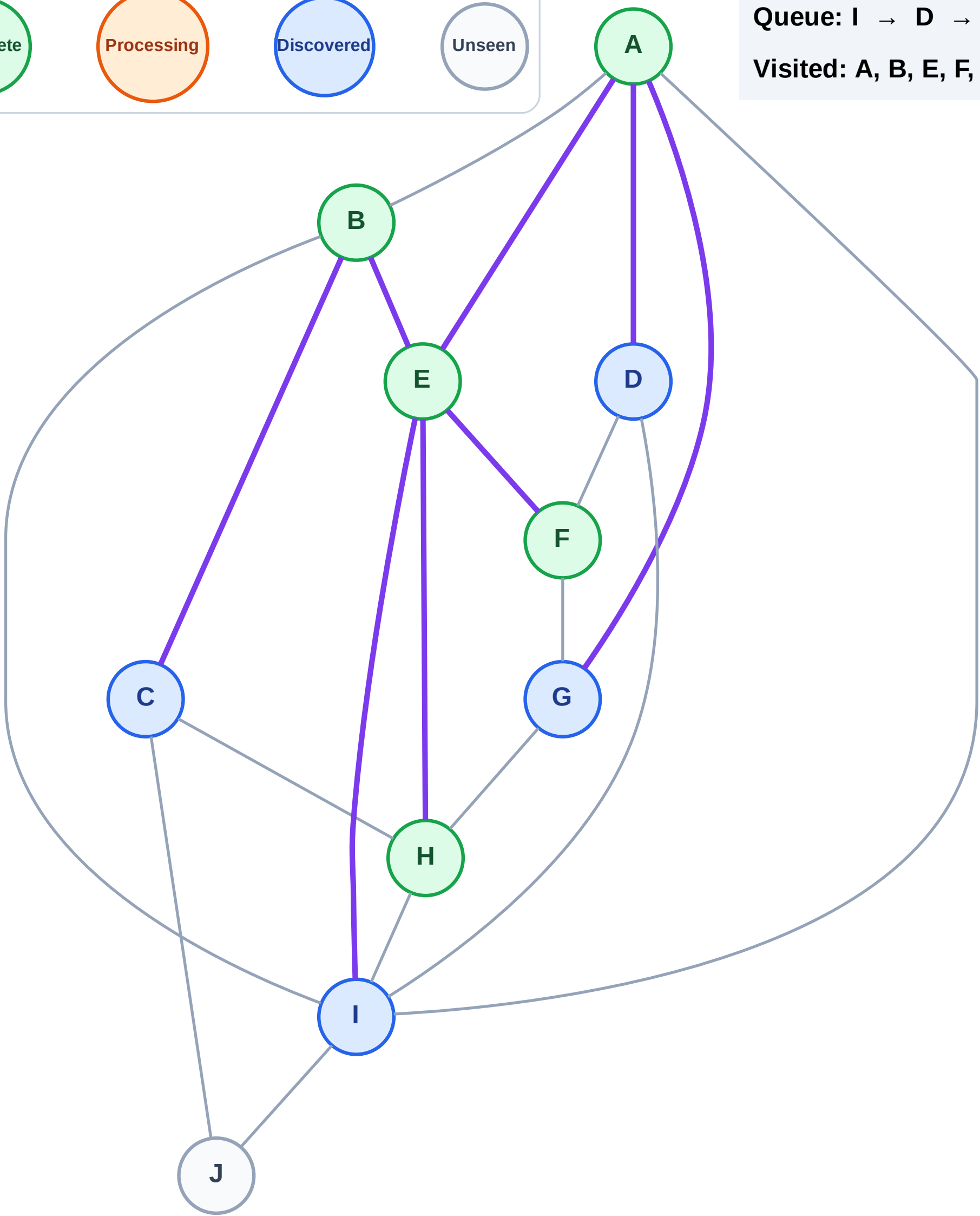
Step 11: No new neighbors from H

Legend



Queue: I → D → G → C

Visited: A, B, E, F, H



BFS Traversal

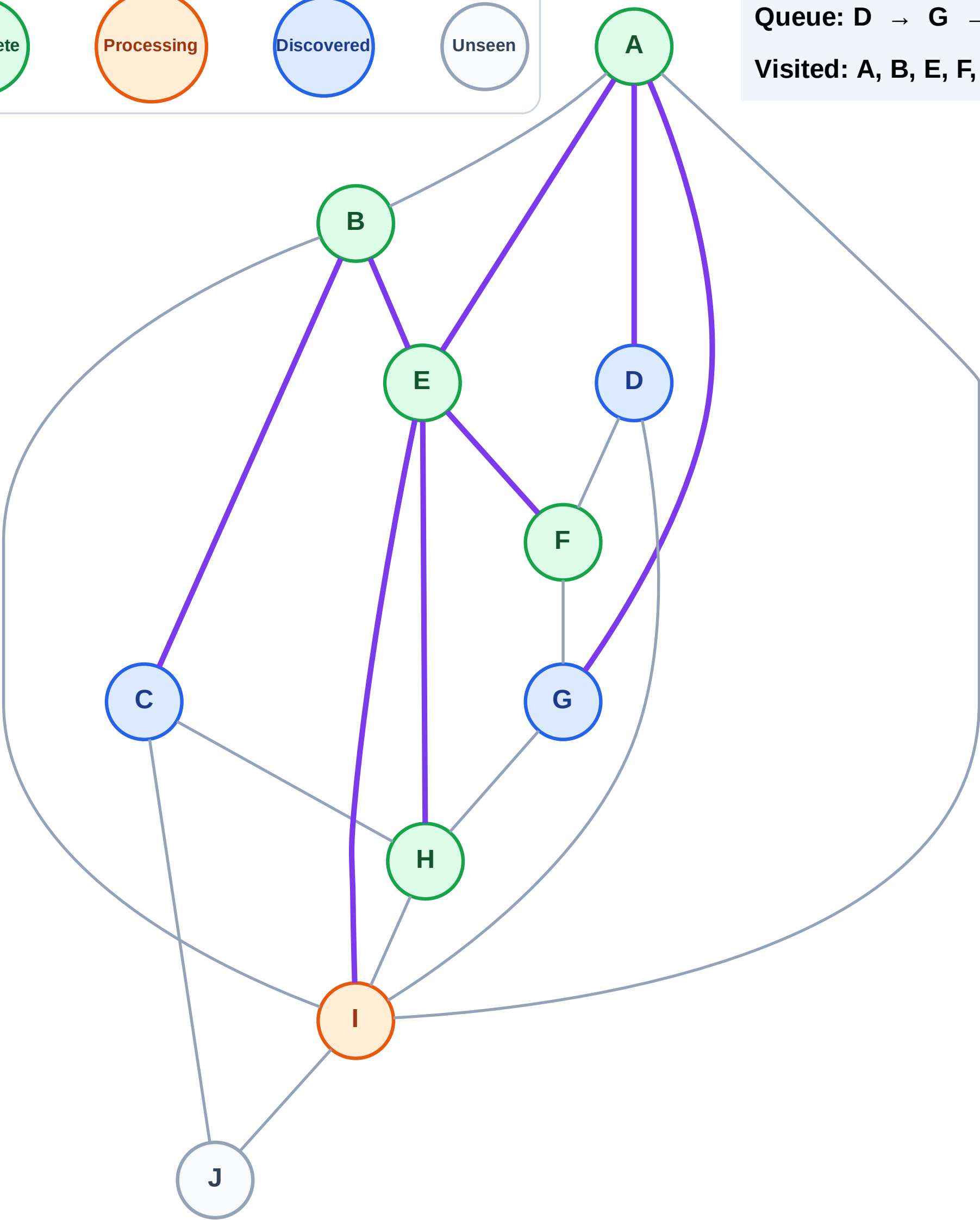
Step 12: Process node I

Legend



Queue: D → G → C

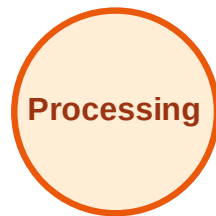
Visited: A, B, E, F, H



BFS Traversal

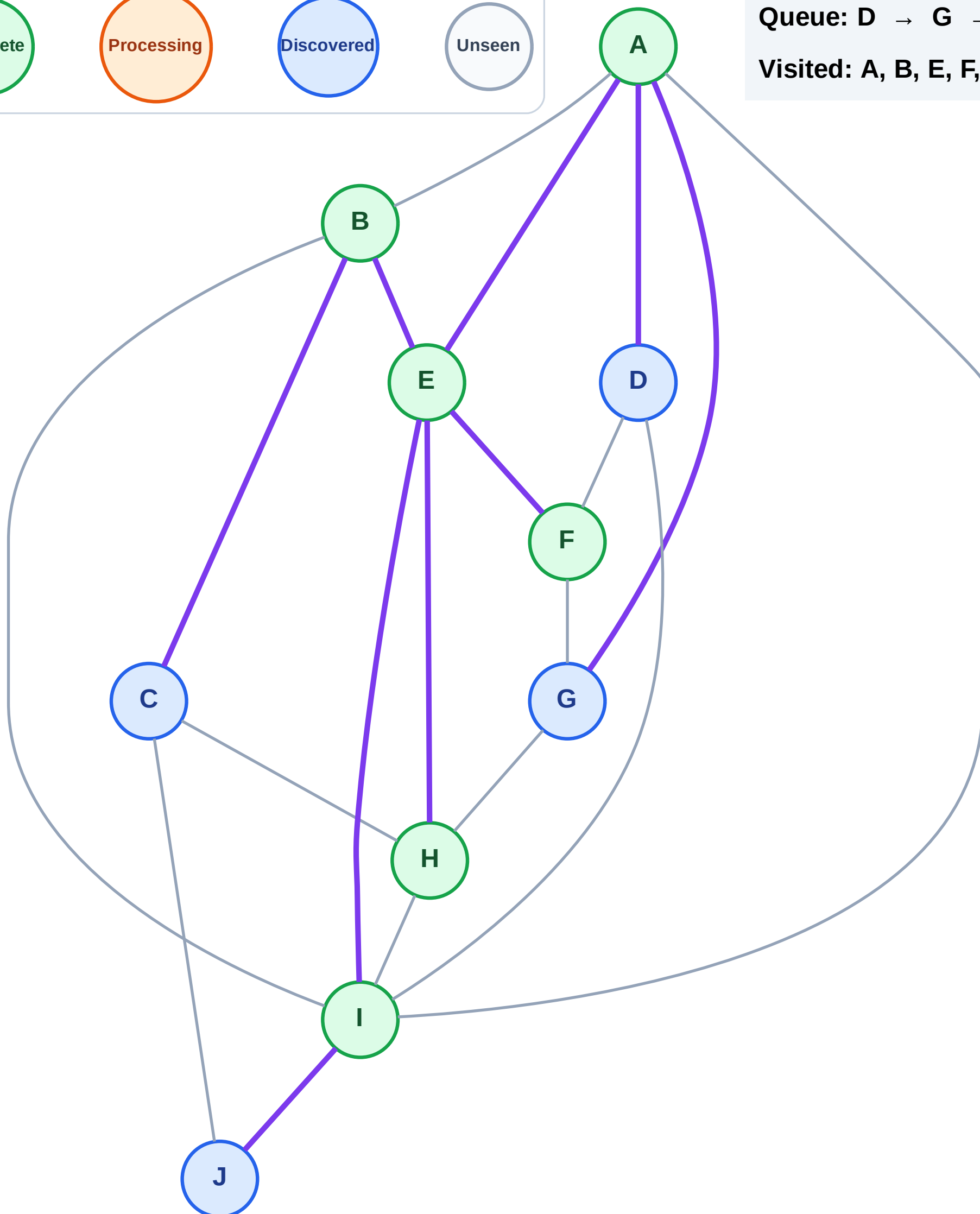
Step 13: Discover J from I

Legend



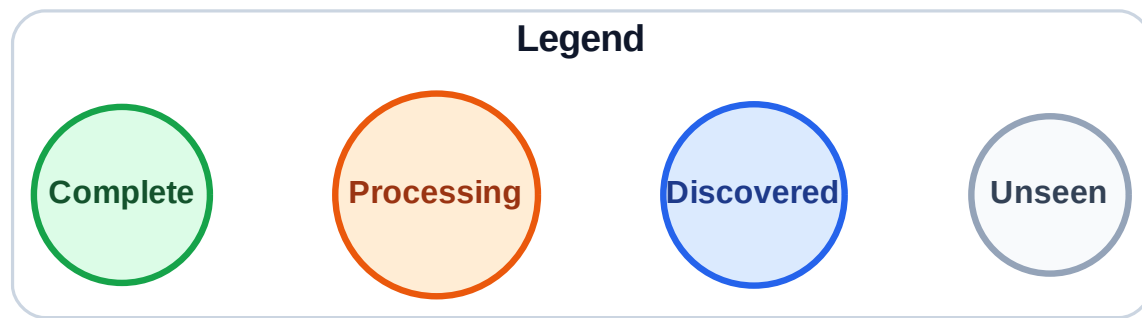
Queue: D → G → C → J

Visited: A, B, E, F, H, I

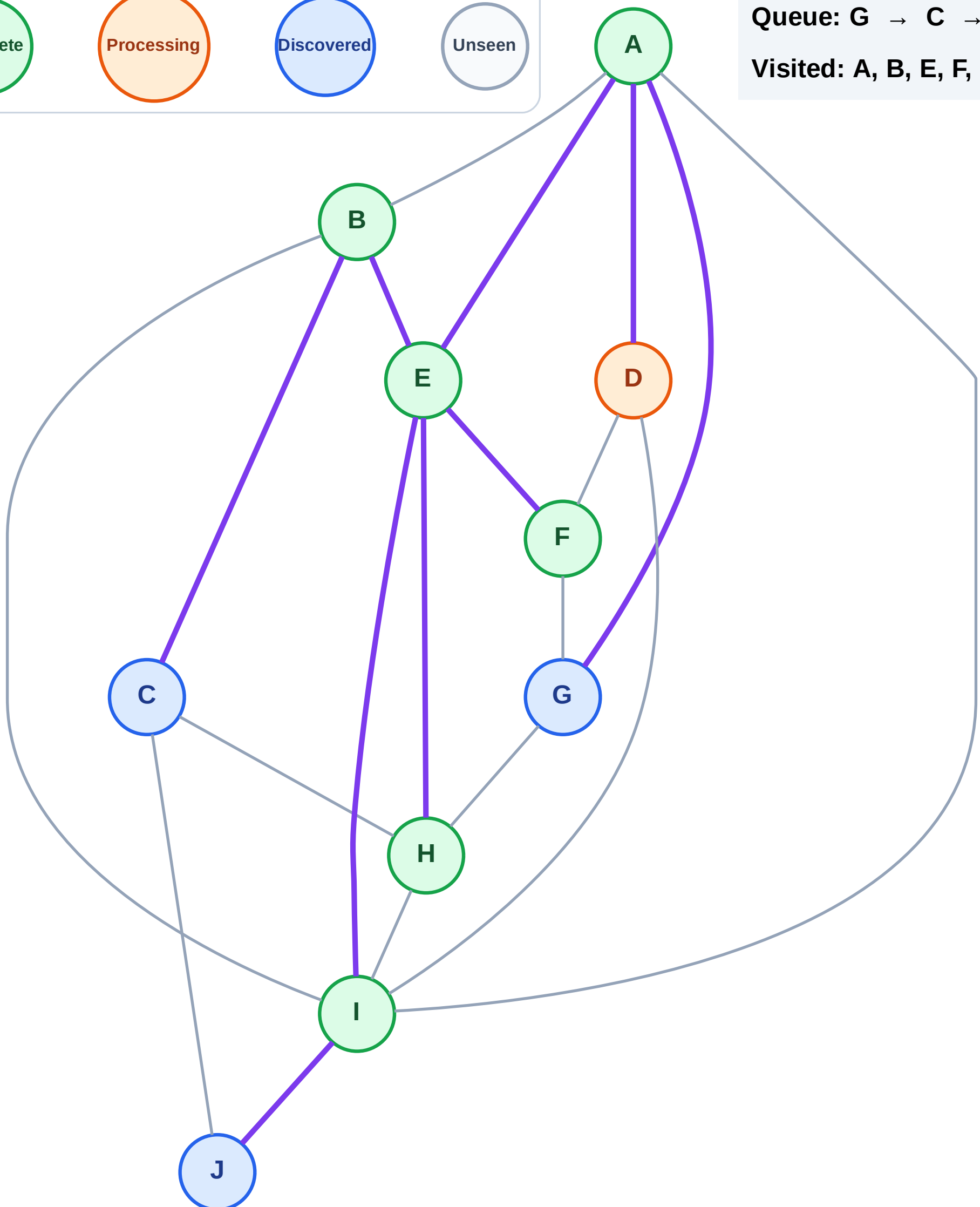


BFS Traversal

Step 14: Process node D



Queue: G → C → J
Visited: A, B, E, F, H, I



BFS Traversal

Step 15: No new neighbors from D

Legend

Complete

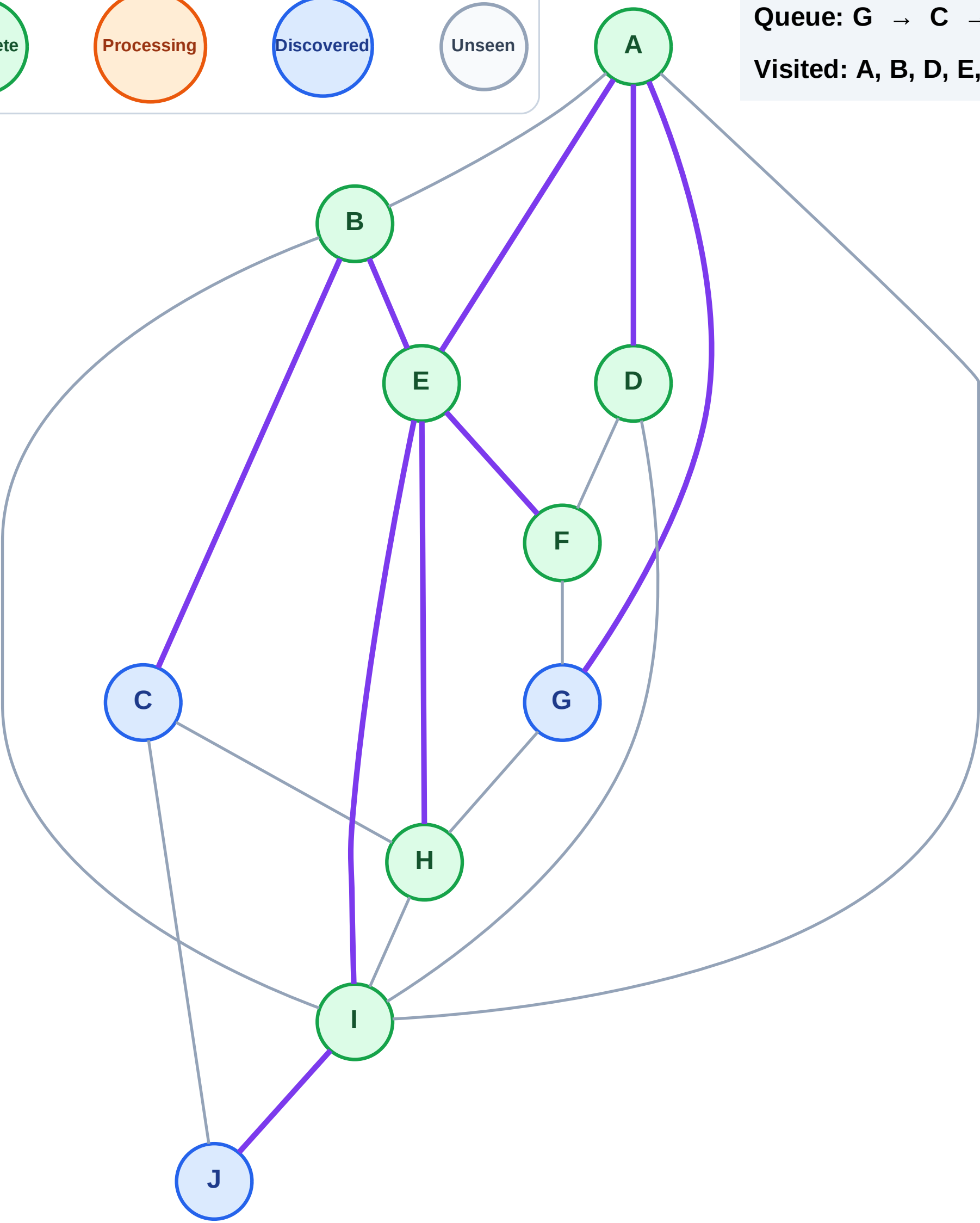
Processing

Discovered

Unseen

Queue: G → C → J

Visited: A, B, D, E, F, H, I



BFS Traversal

Step 16: Process node G

Legend

Complete

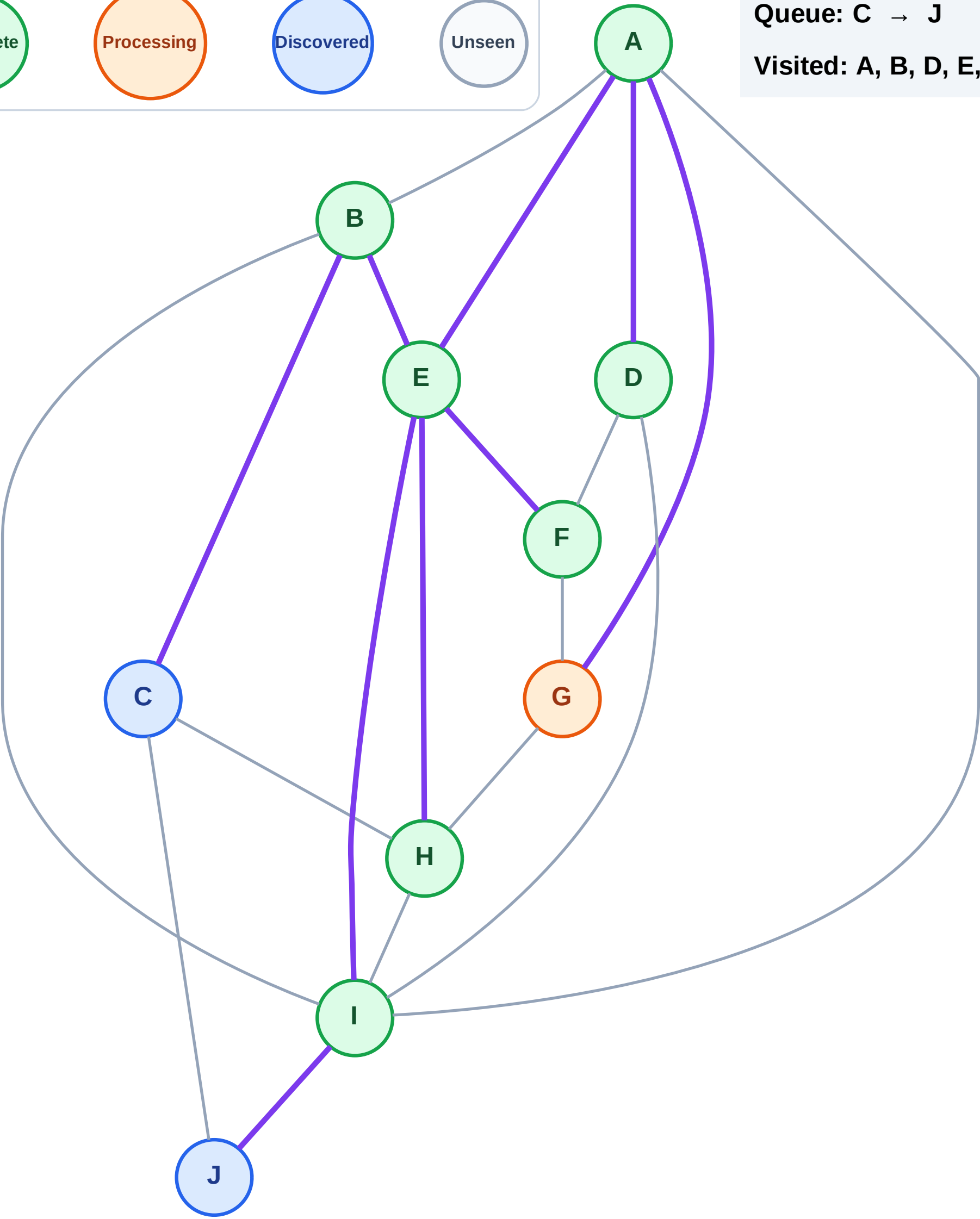
Processing

Discovered

Unseen

Queue: C → J

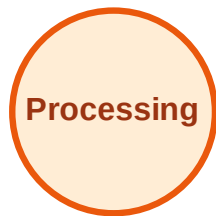
Visited: A, B, D, E, F, H, I



BFS Traversal

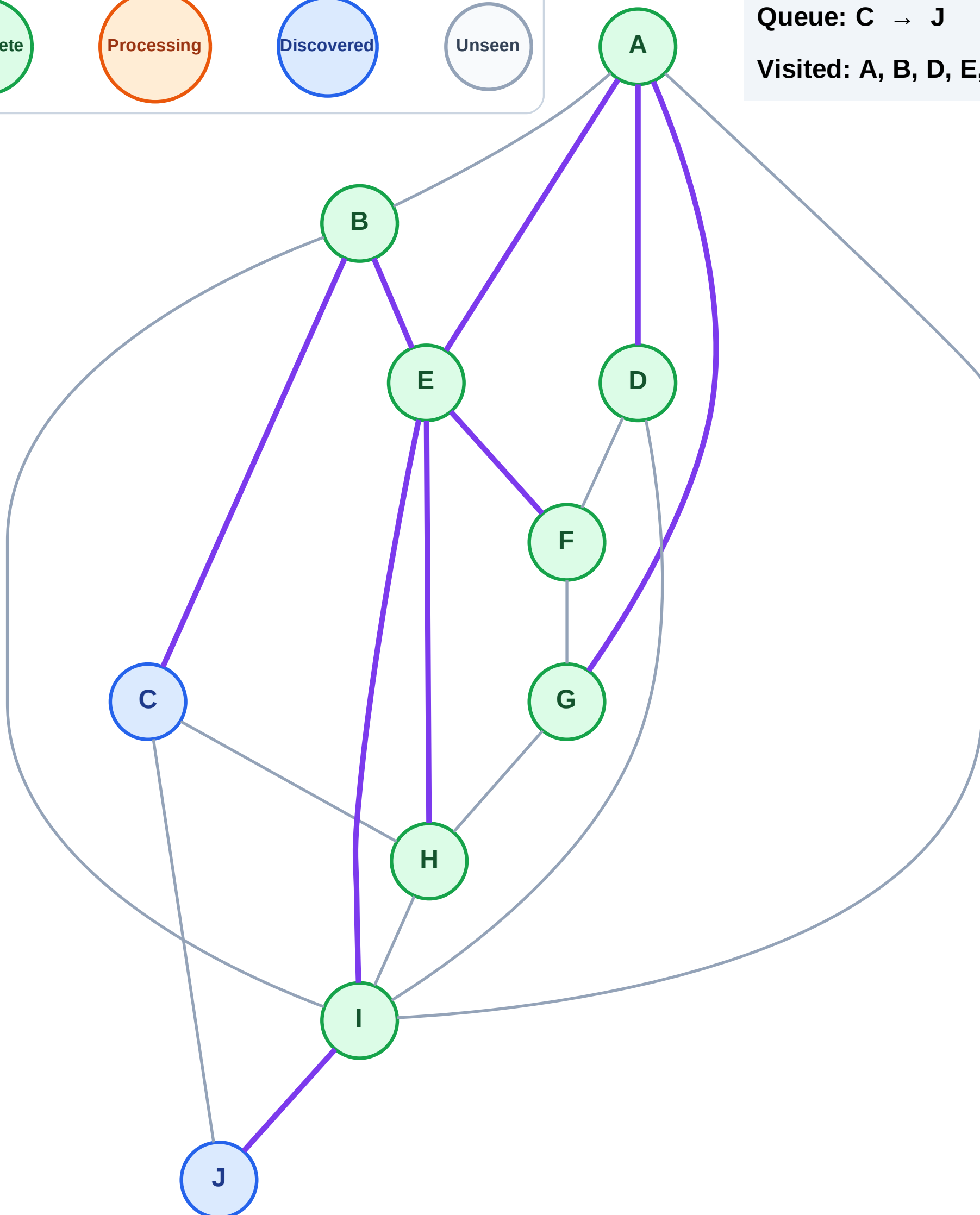
Step 17: No new neighbors from G

Legend



Queue: C → J

Visited: A, B, D, E, F, G, H, I



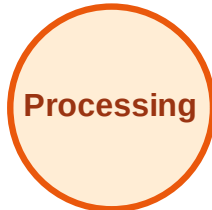
BFS Traversal

Step 18: Process node C

Legend



Complete



Processing



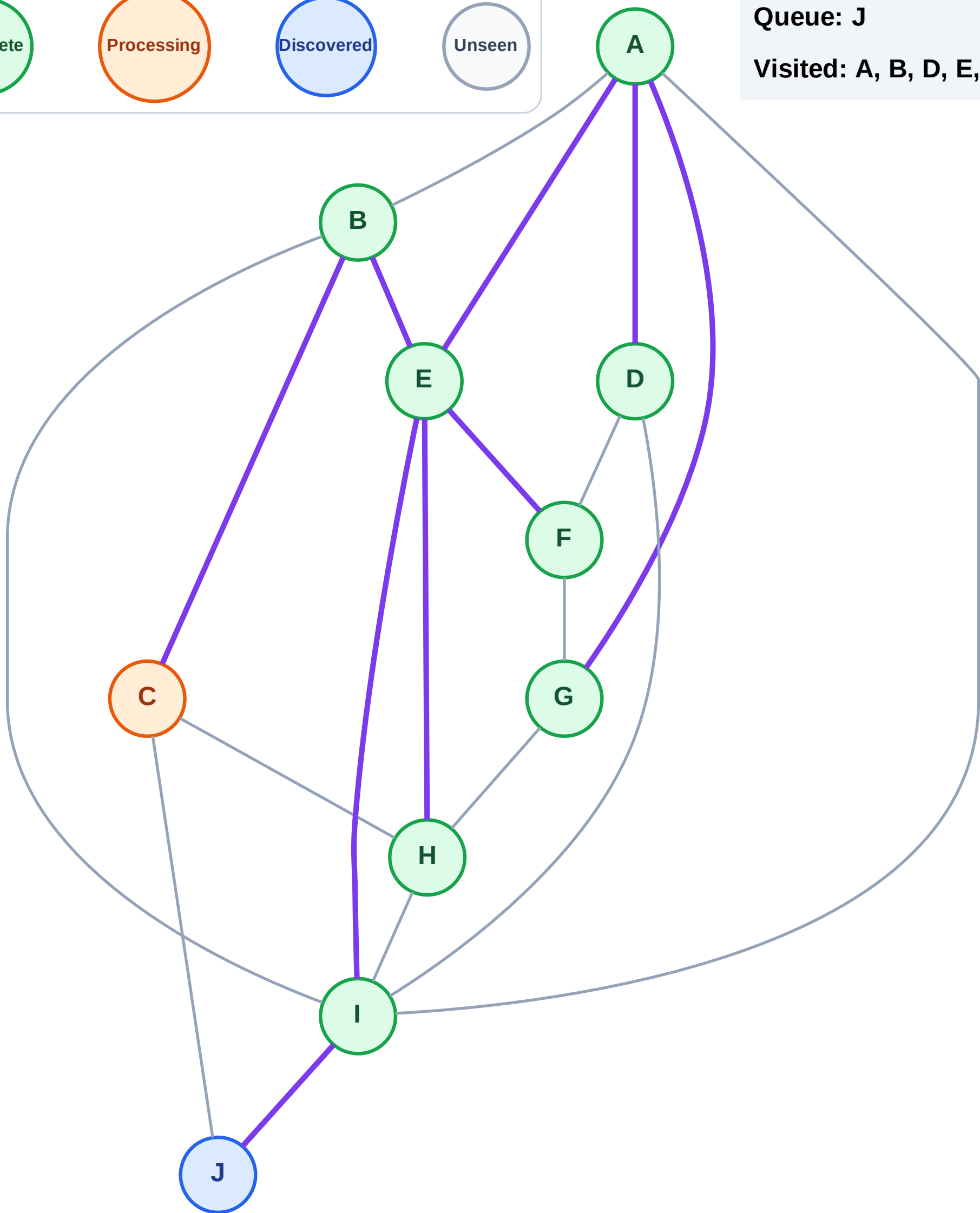
Discovered



Unseen

Queue: J

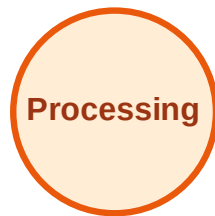
Visited: A, B, D, E, F, G, H, I



BFS Traversal

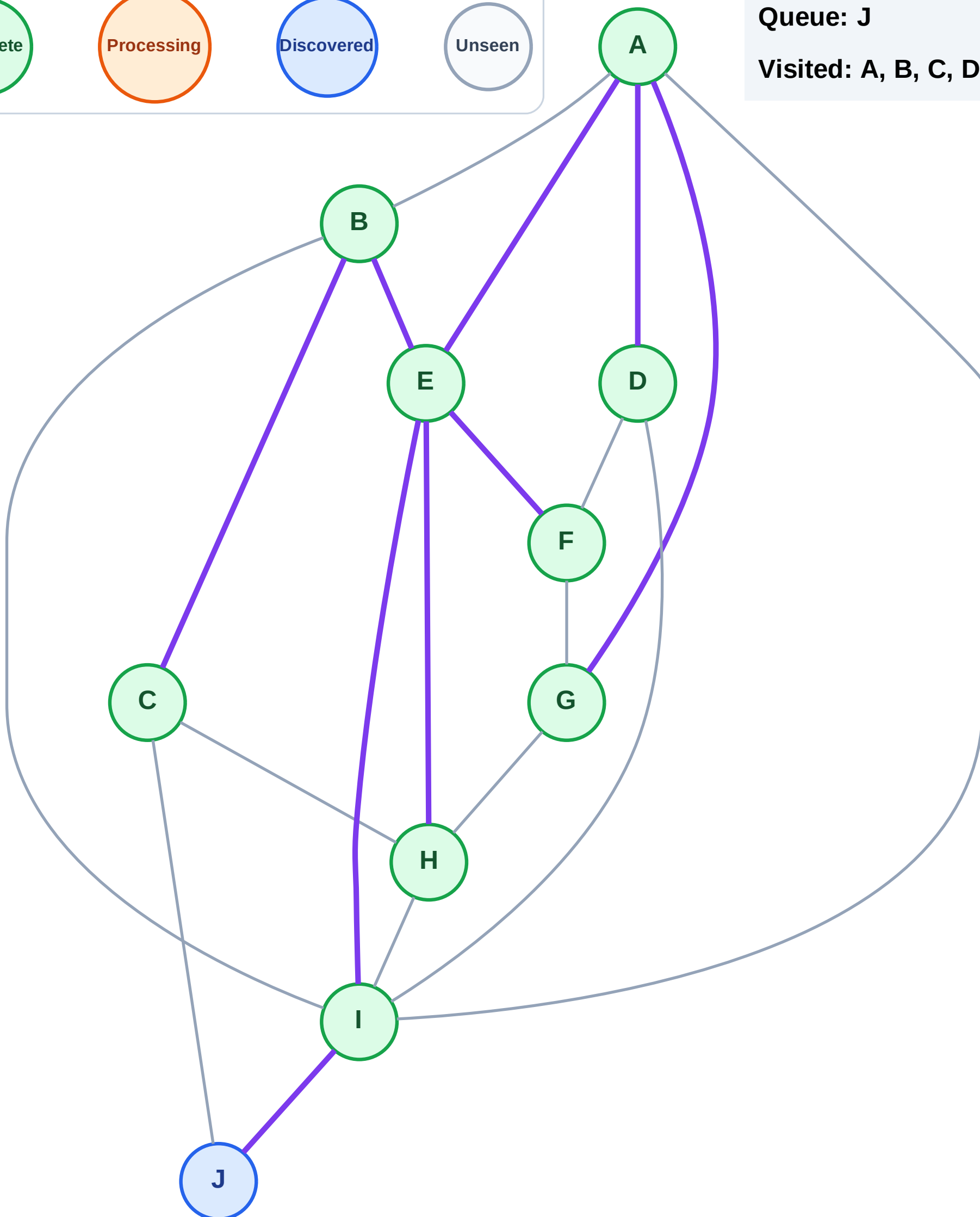
Step 19: No new neighbors from C

Legend



Queue: J

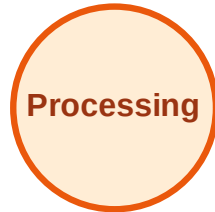
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

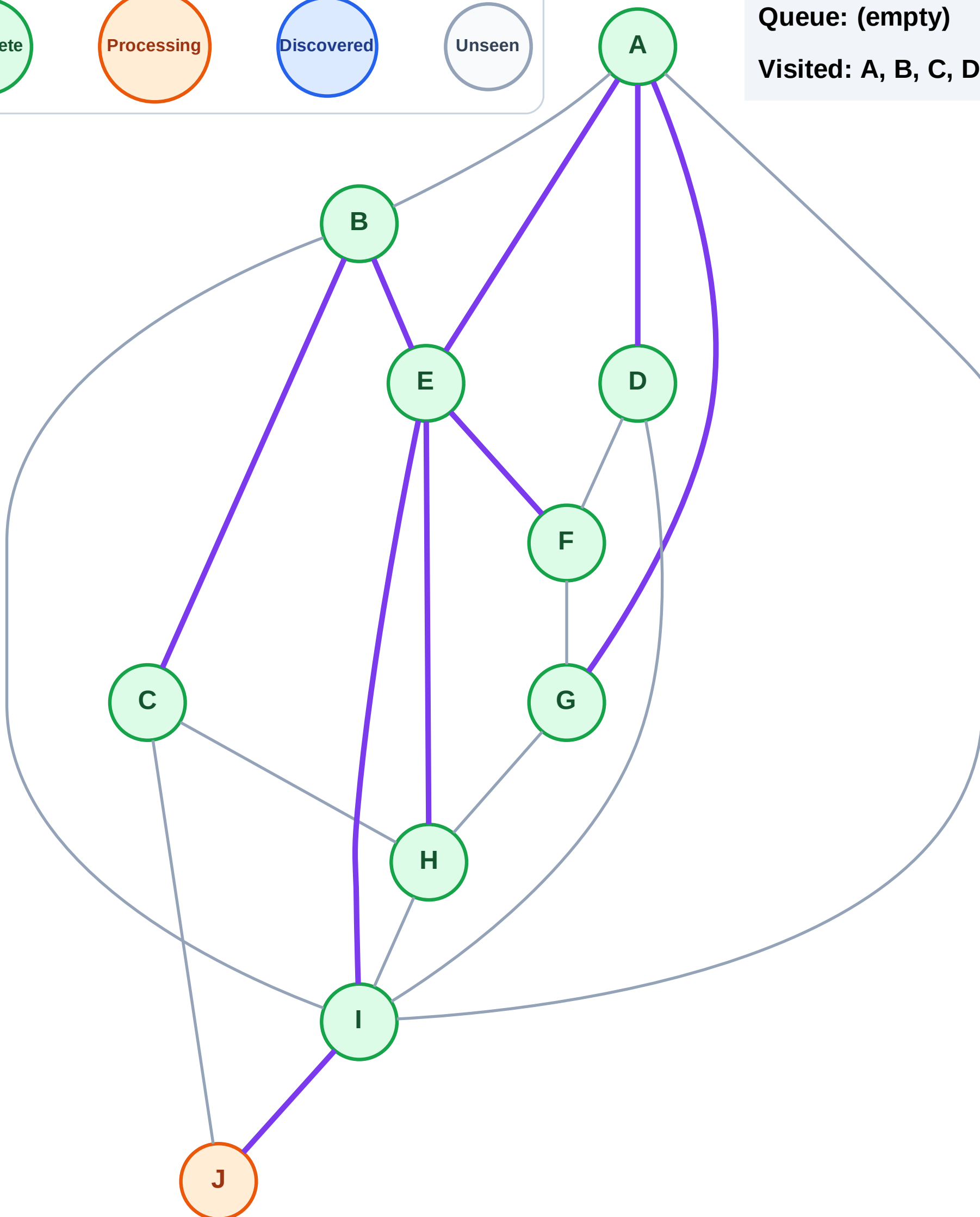
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

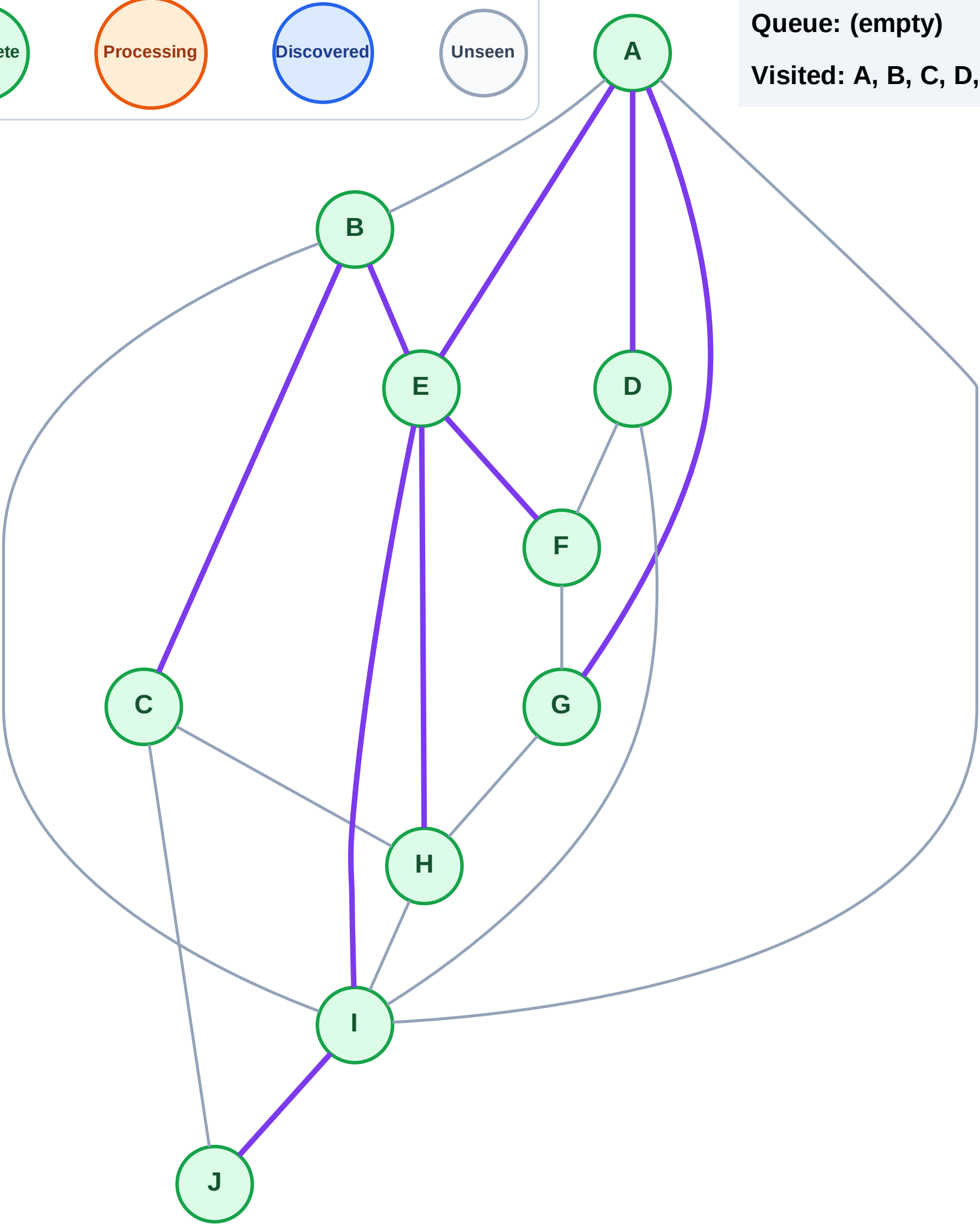
Step 21: No new neighbors from J

Legend



Queue: (empty)

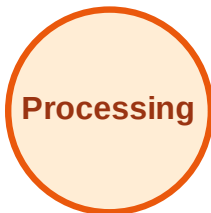
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

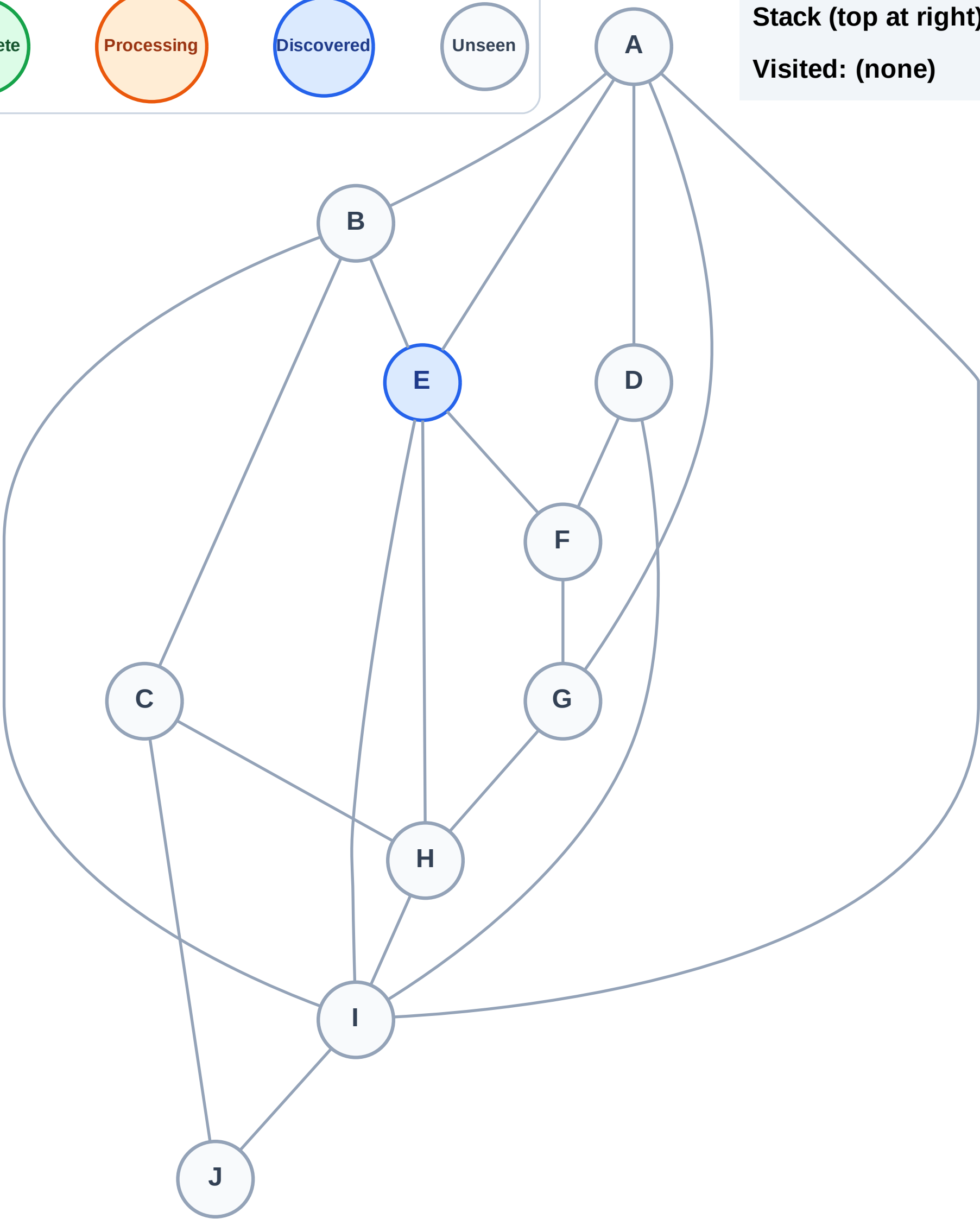
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

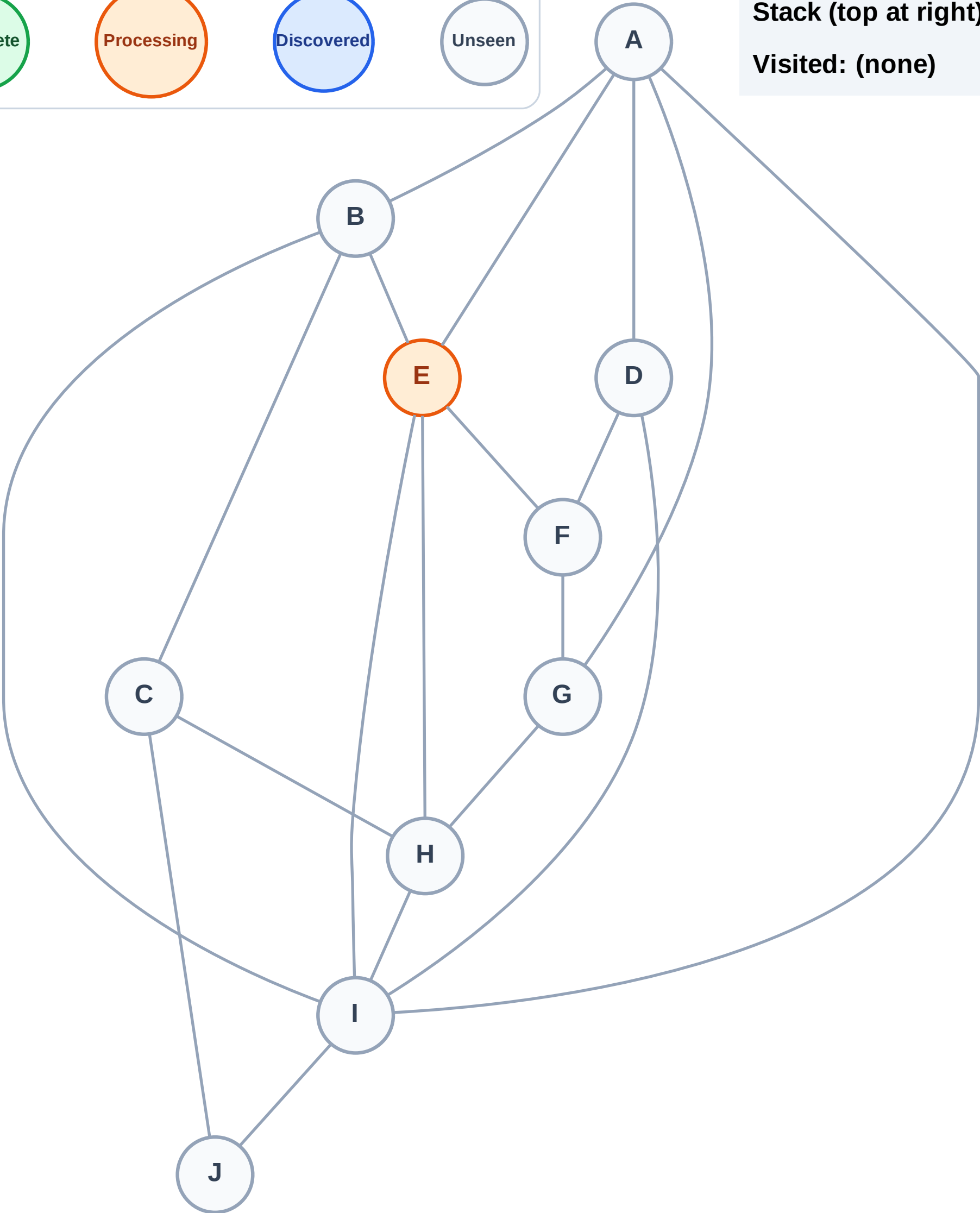
Complete

Processing

Discovered

Unseen

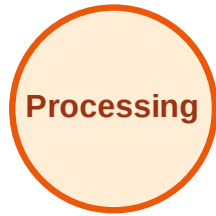
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

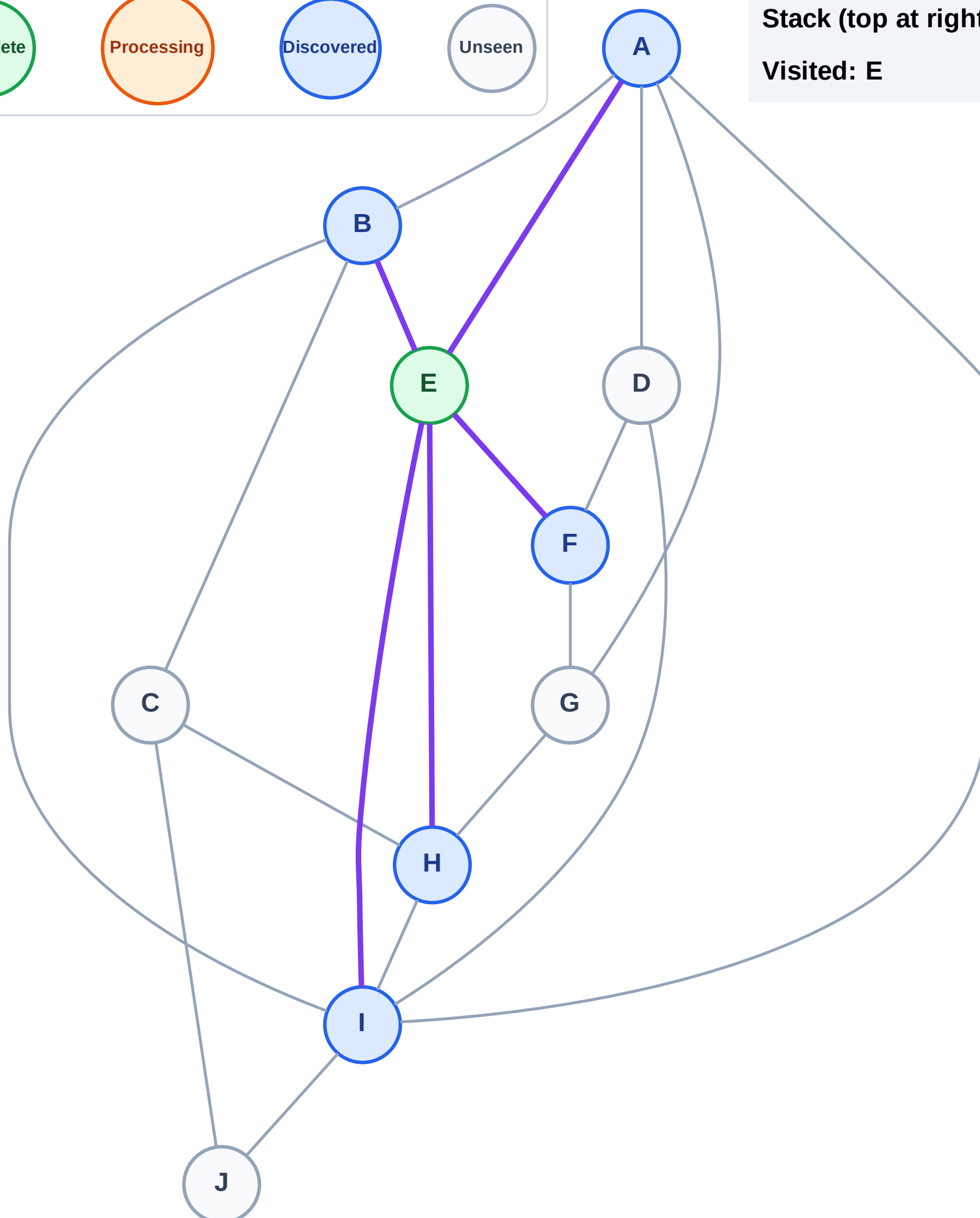
Step 3: Discover I, H, F, B, A from E

Legend



Stack (top at right): I | H | F | B | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

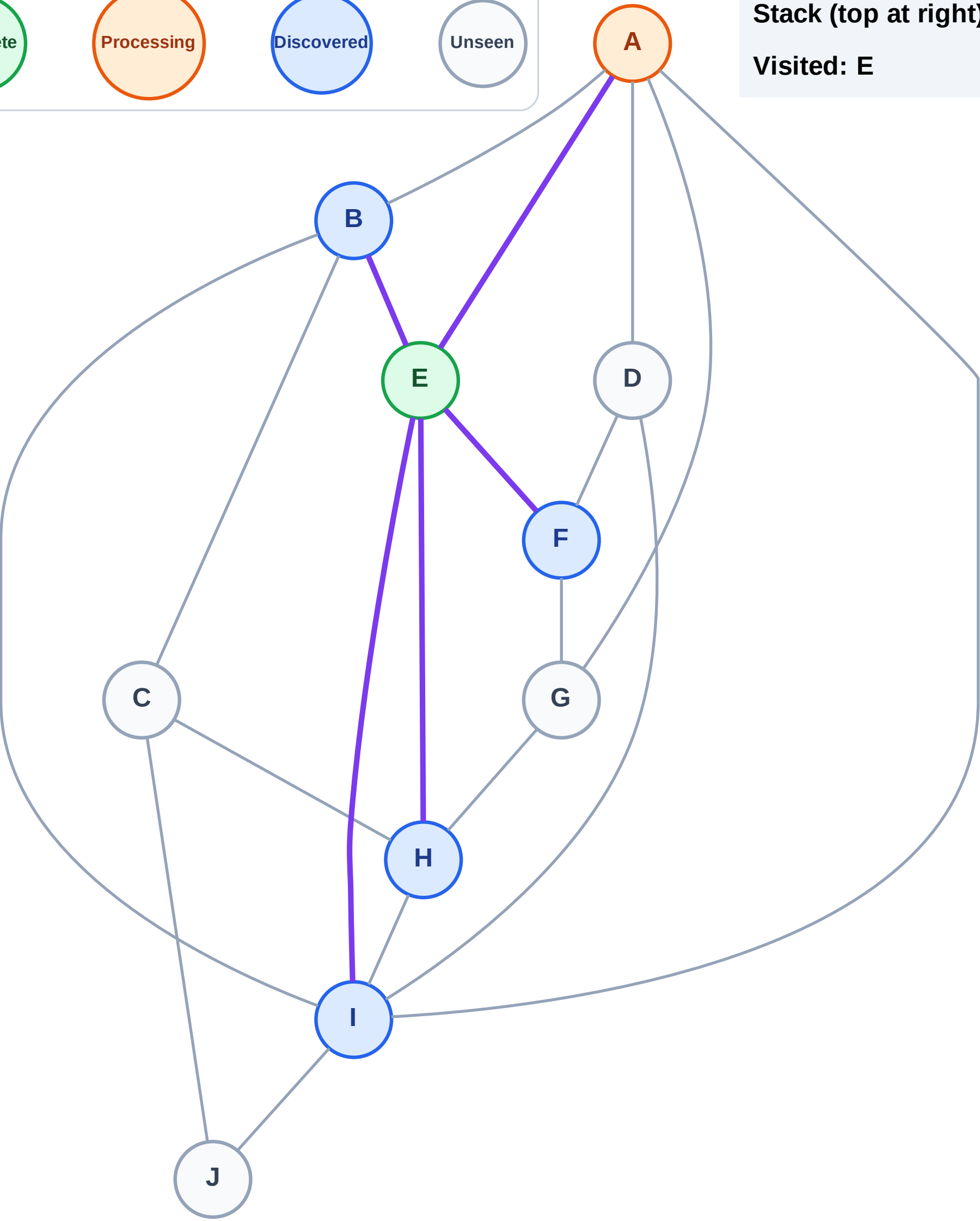
Processing

Discovered

Unseen

Stack (top at right): I | H | F | B

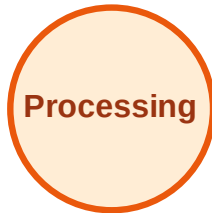
Visited: E



DFS Traversal

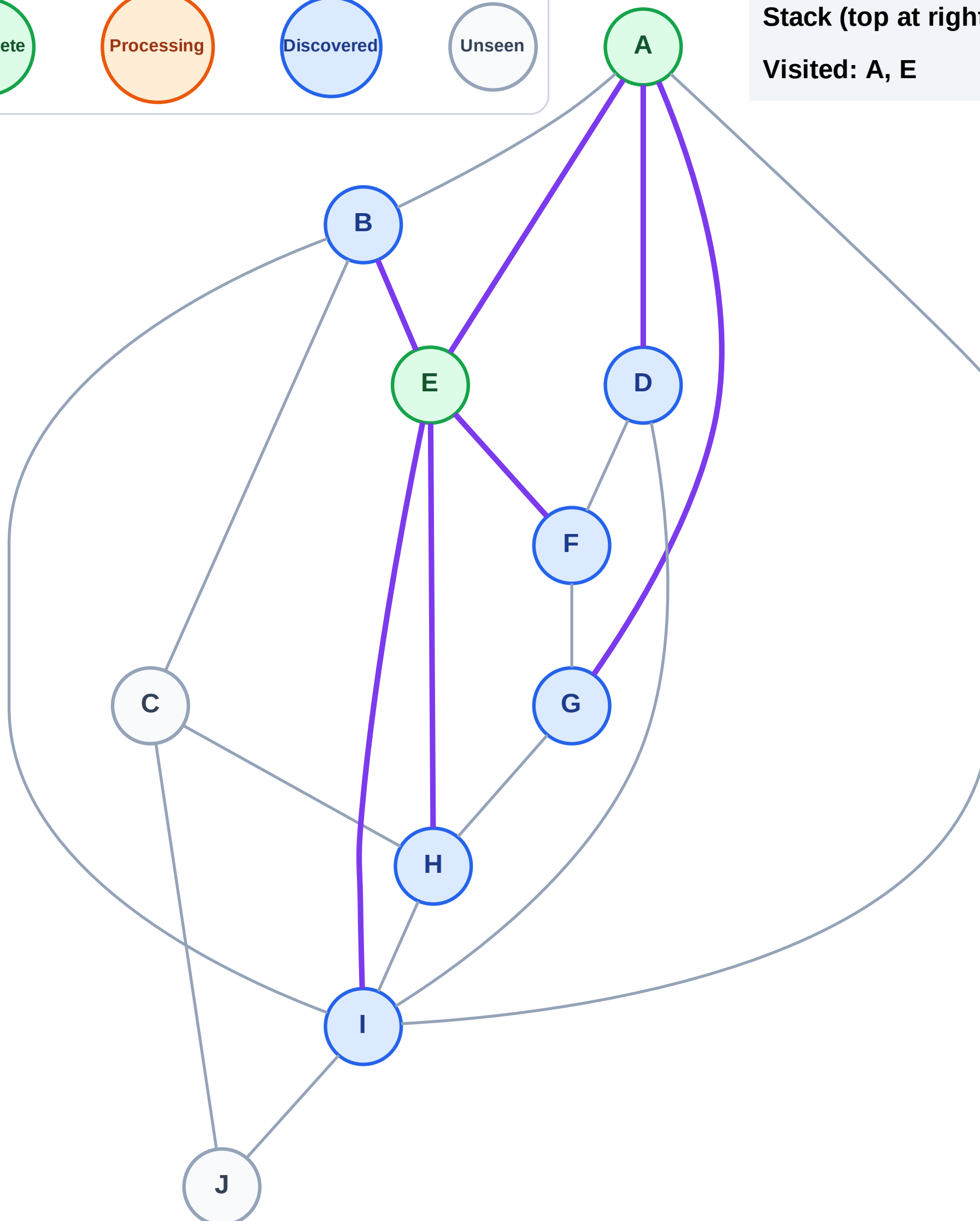
Step 5: Discover G, D from A

Legend



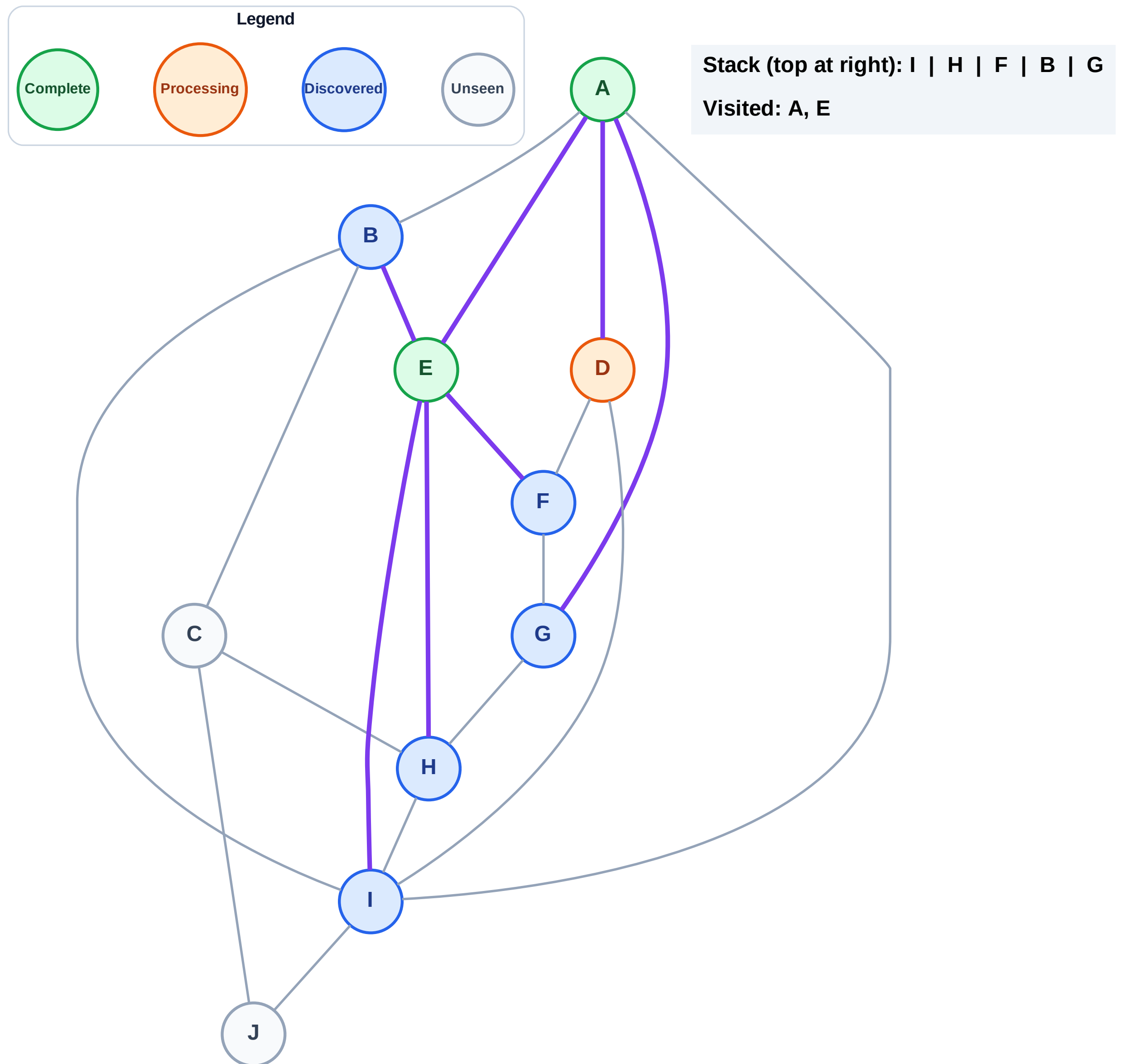
Stack (top at right): I | H | F | B | G | D

Visited: A, E



Step 6: Process node D

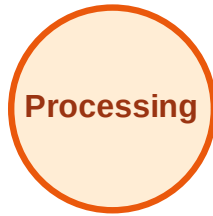
Step 6: Process node D



DFS Traversal

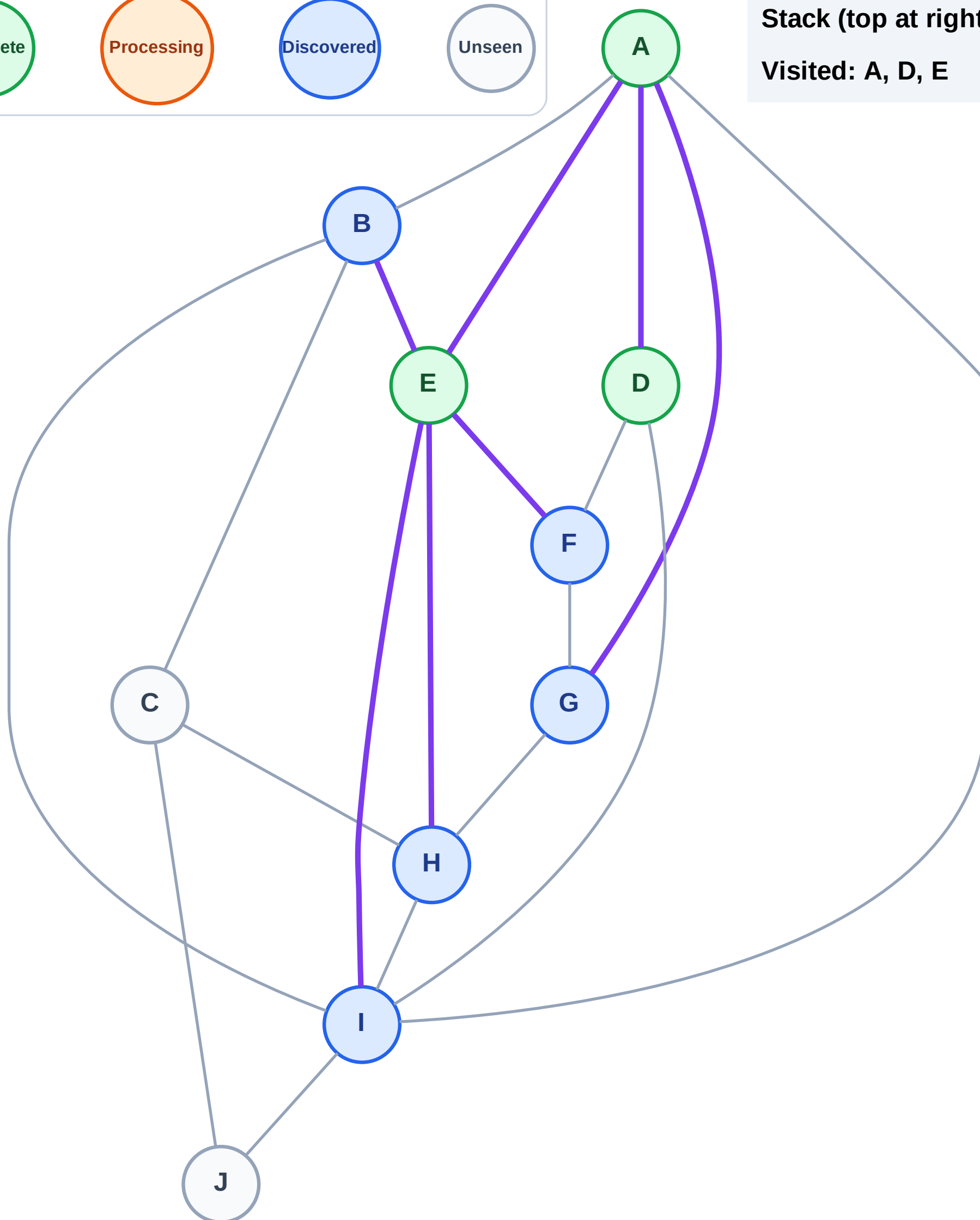
Step 7: No new neighbors from D

Legend



Stack (top at right): I | H | F | B | G

Visited: A, D, E



DFS Traversal

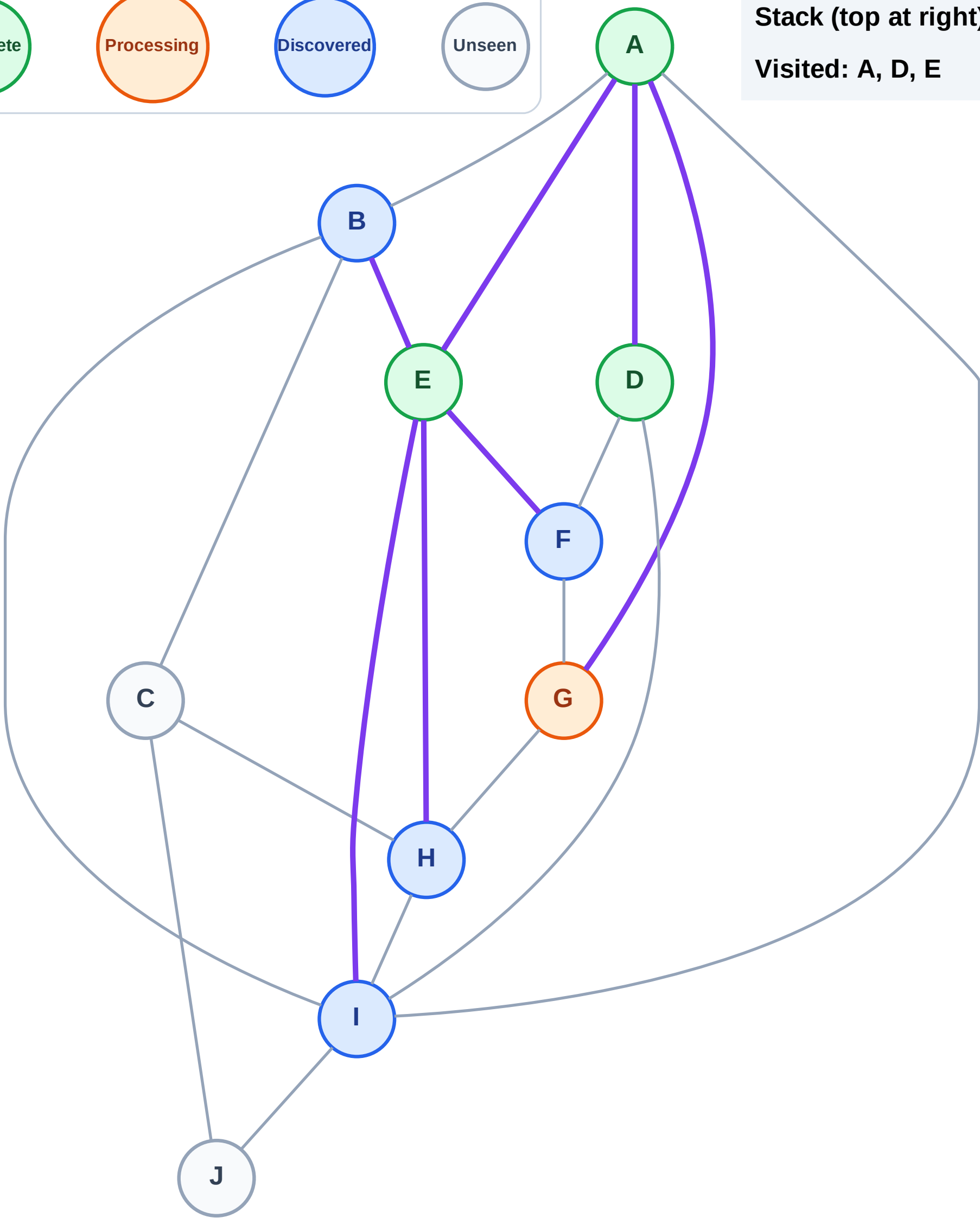
Step 8: Process node G

Legend



Stack (top at right): I | H | F | B

Visited: A, D, E



DFS Traversal

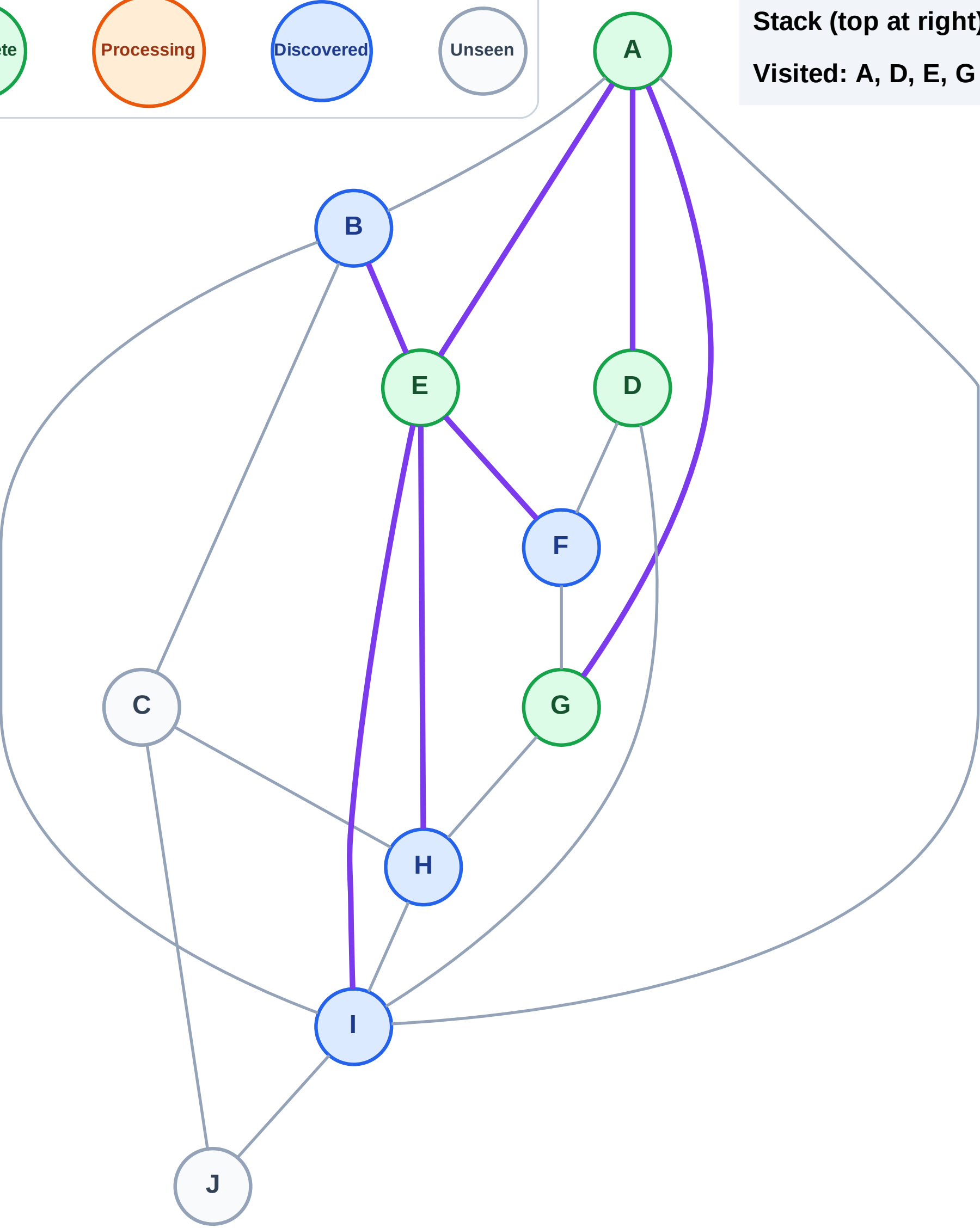
Step 9: No new neighbors from G

Legend



Stack (top at right): I | H | F | B

Visited: A, D, E, G



DFS Traversal

Step 10: Process node B

Legend

Complete

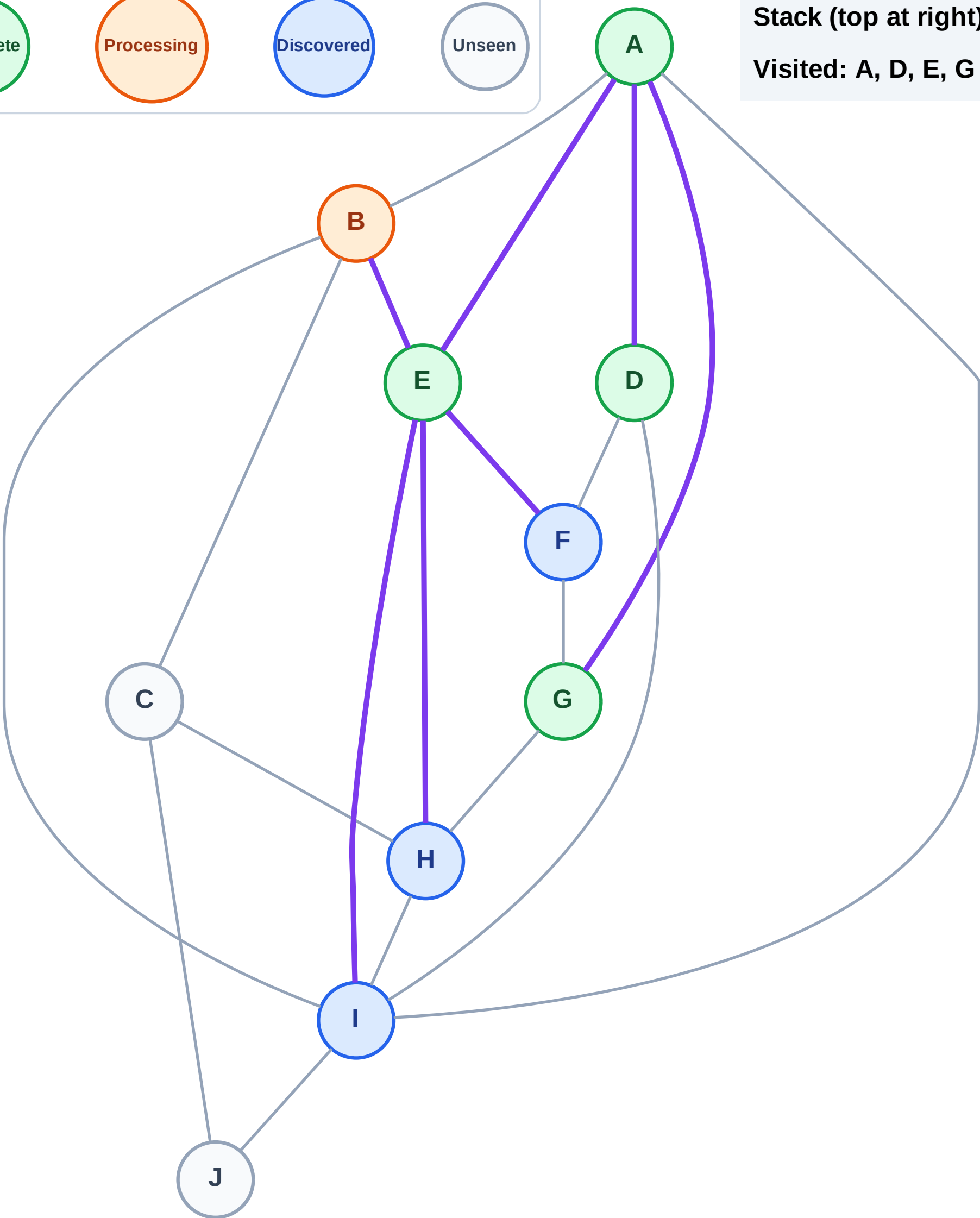
Processing

Discovered

Unseen

Stack (top at right): I | H | F

Visited: A, D, E, G



DFS Traversal

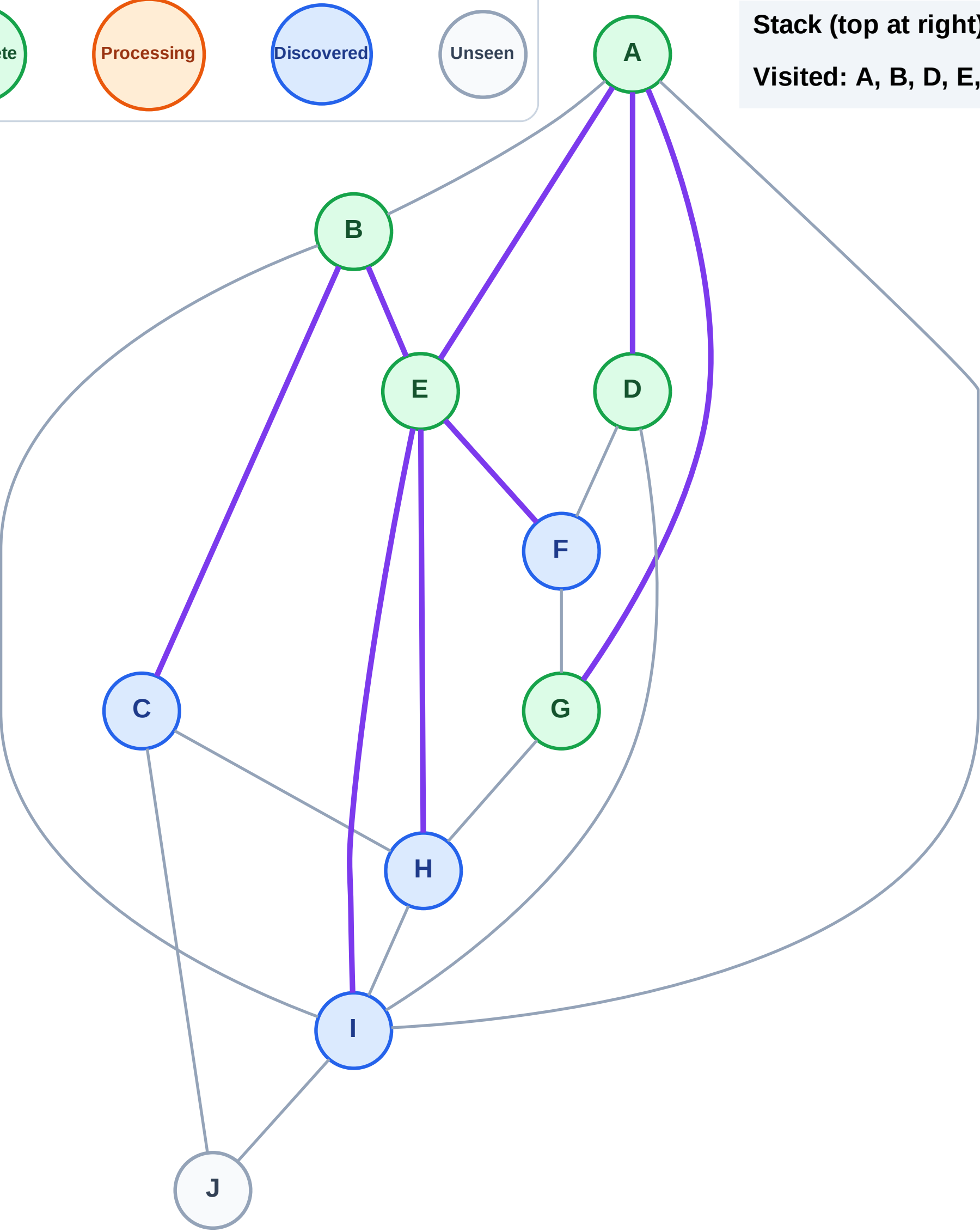
Step 11: Discover C from B

Legend



Stack (top at right): I | H | F | C

Visited: A, B, D, E, G



DFS Traversal

Step 12: Process node C

Legend

Complete

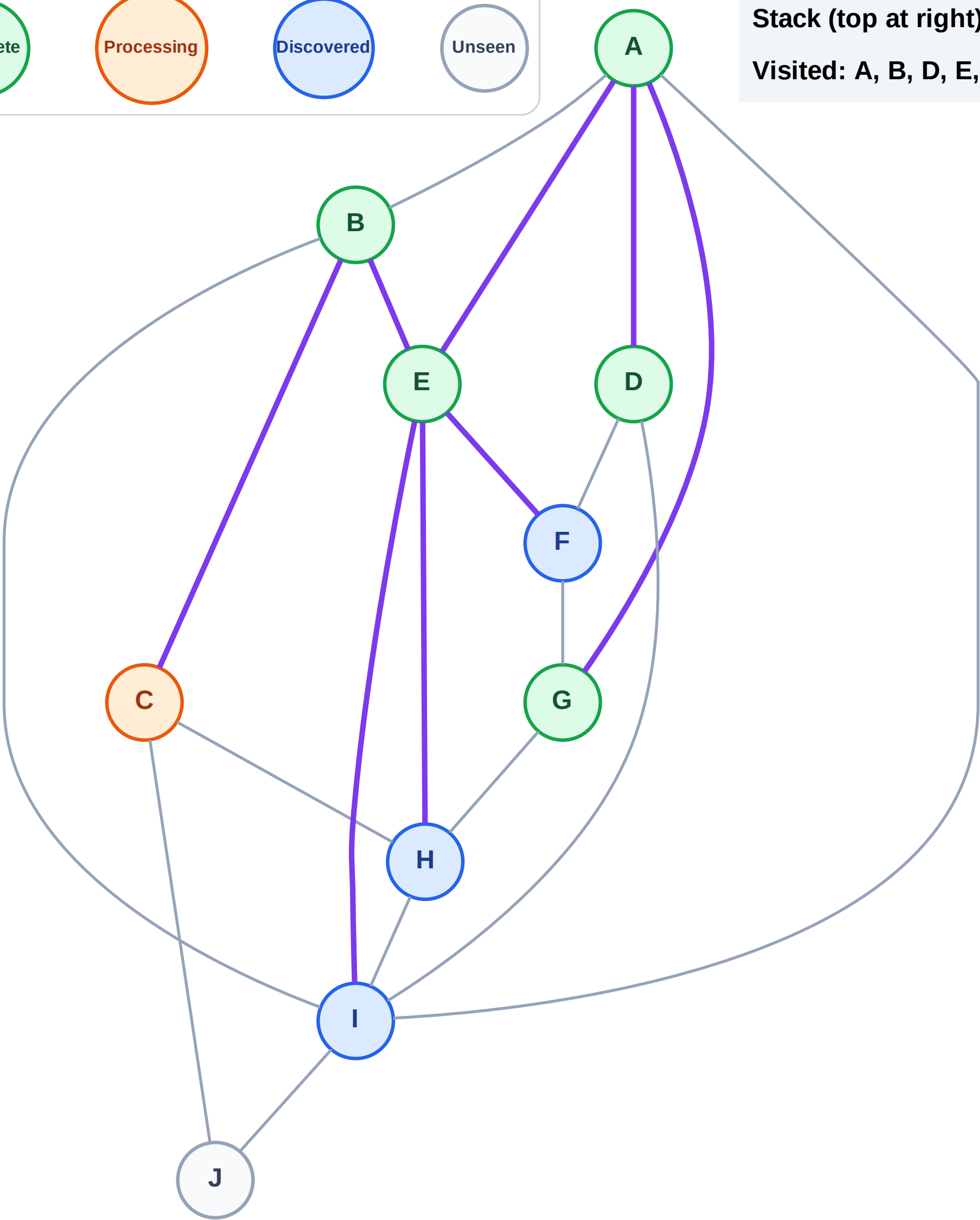
Processing

Discovered

Unseen

Stack (top at right): I | H | F

Visited: A, B, D, E, G



DFS Traversal

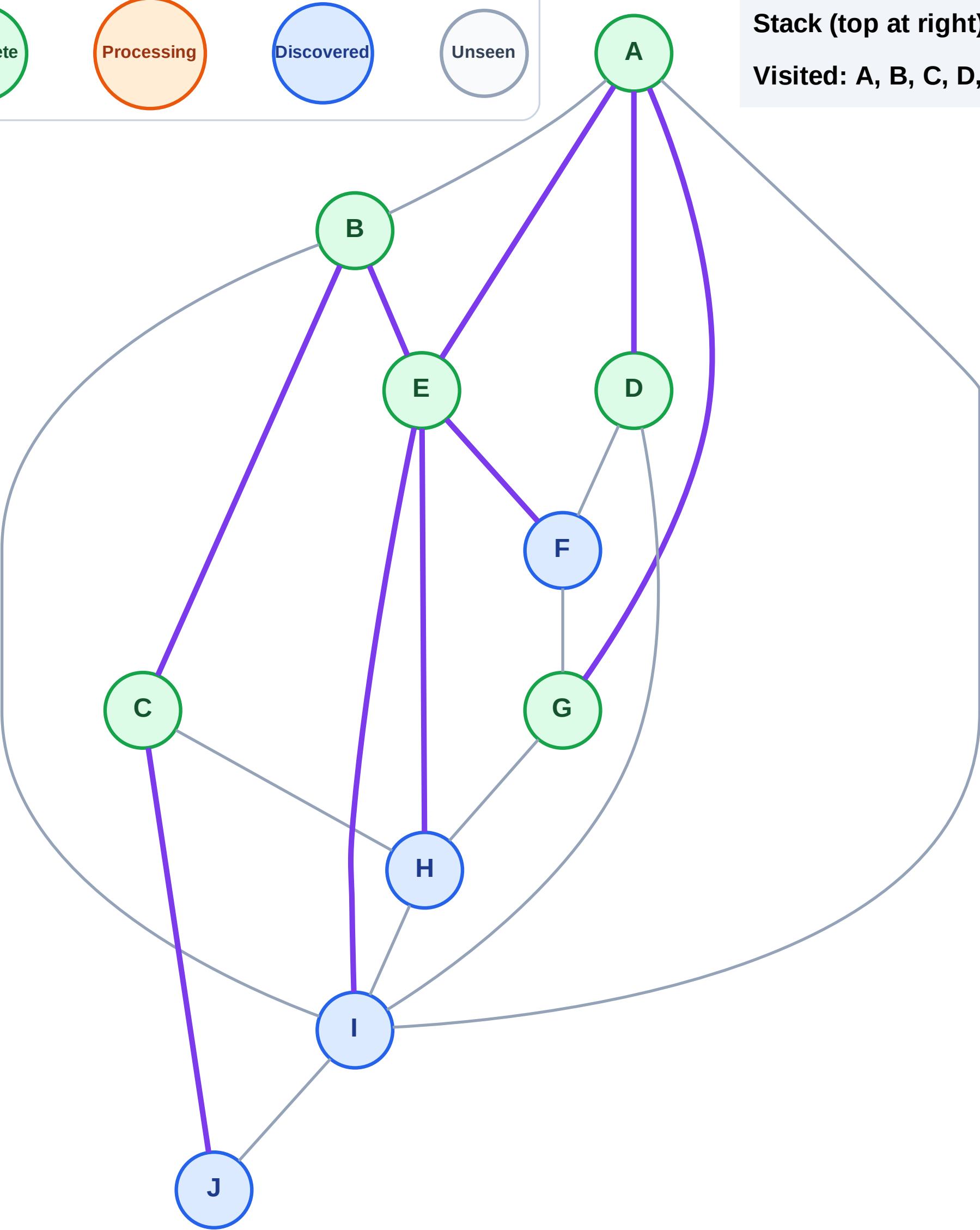
Step 13: Discover J from C

Legend

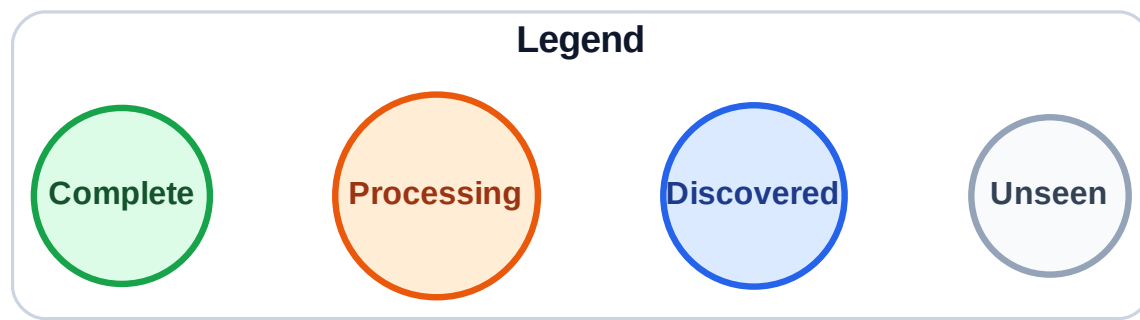


Stack (top at right): I | H | F | J

Visited: A, B, C, D, E, G

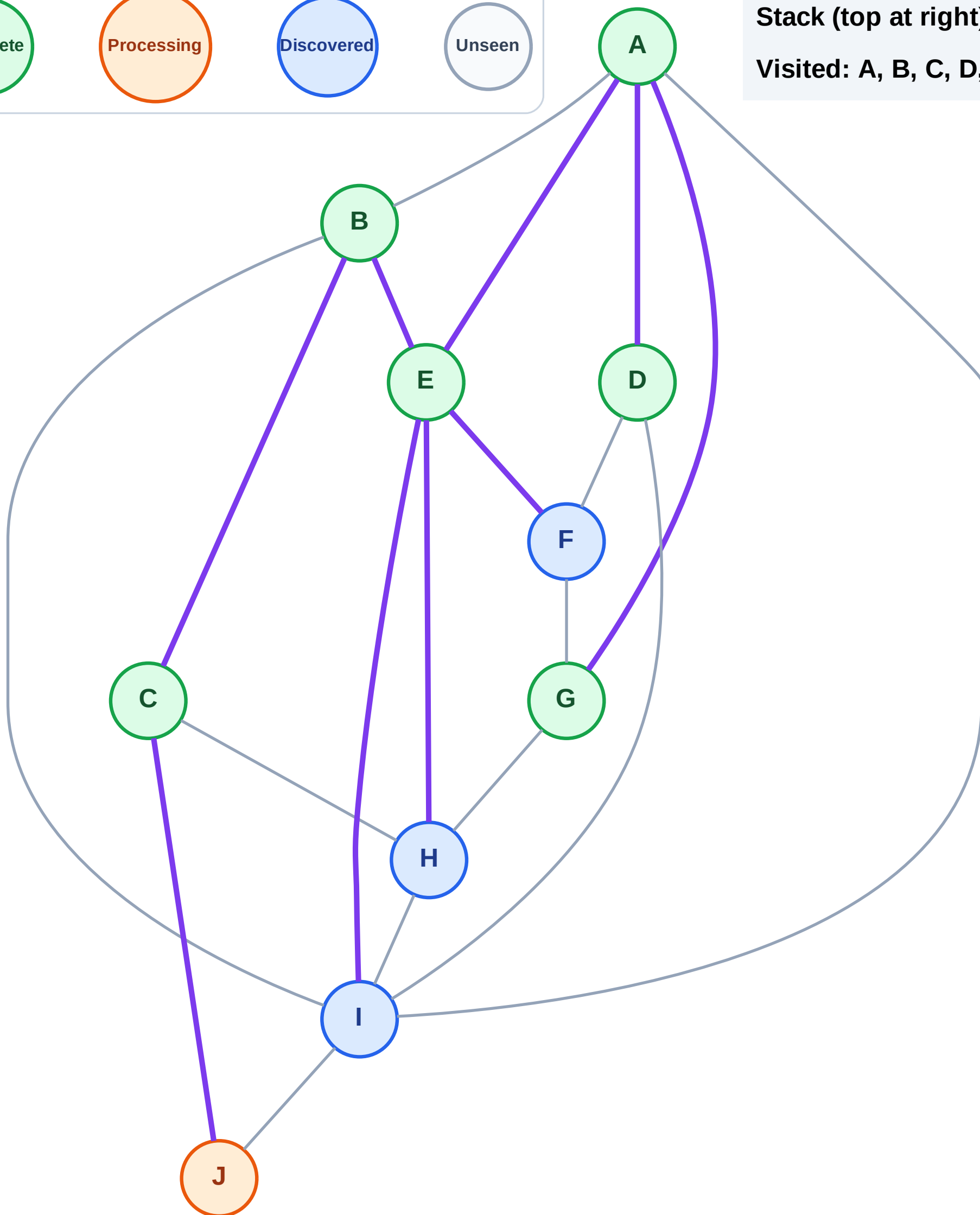


Step 14: Process node J



Stack (top at right): I | H | F

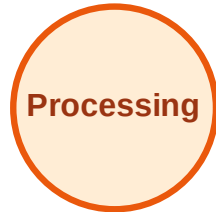
Visited: A, B, C, D, E, G



DFS Traversal

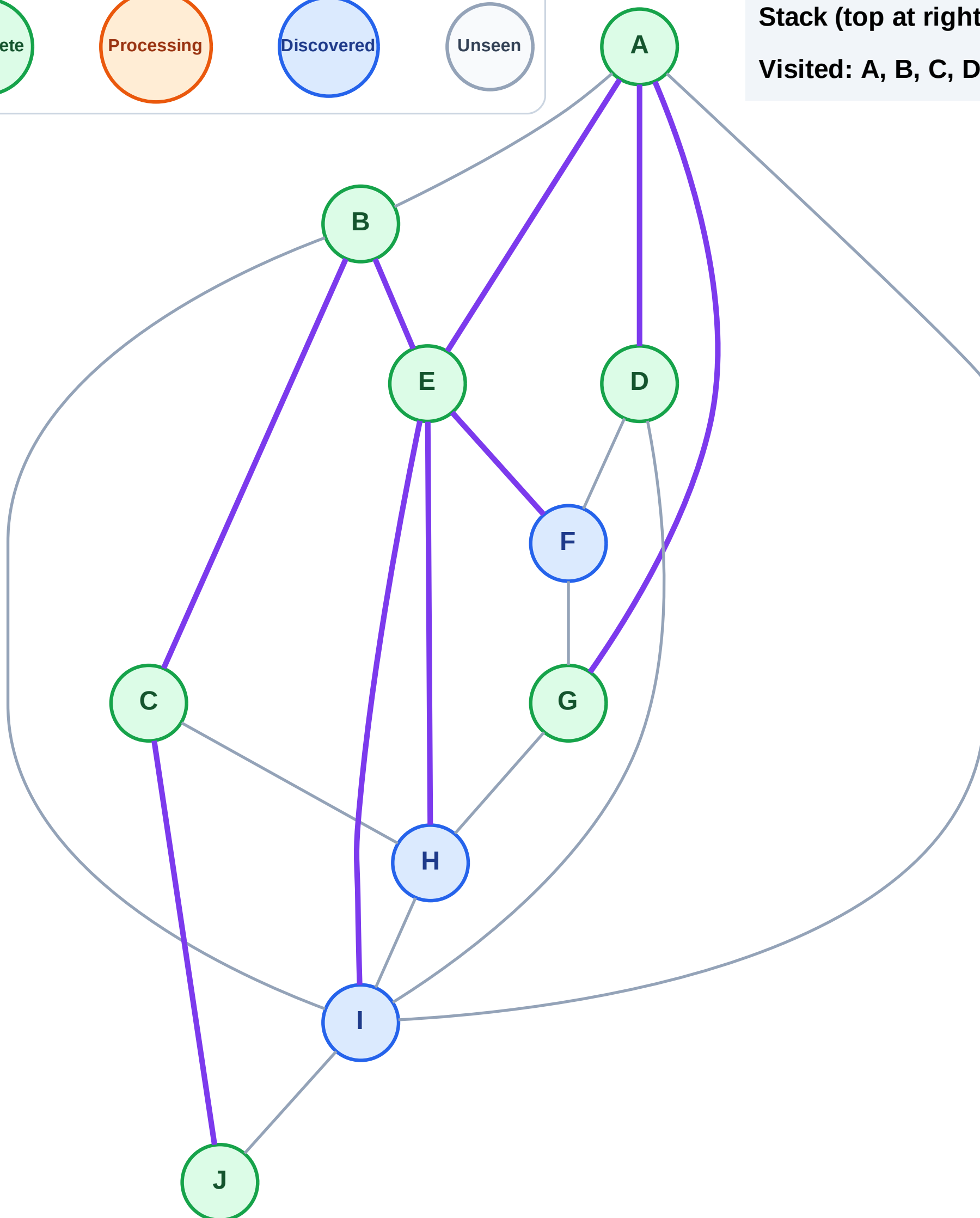
Step 15: No new neighbors from J

Legend



Stack (top at right): I | H | F

Visited: A, B, C, D, E, G, J



DFS Traversal

Step 16: Process node F

Legend

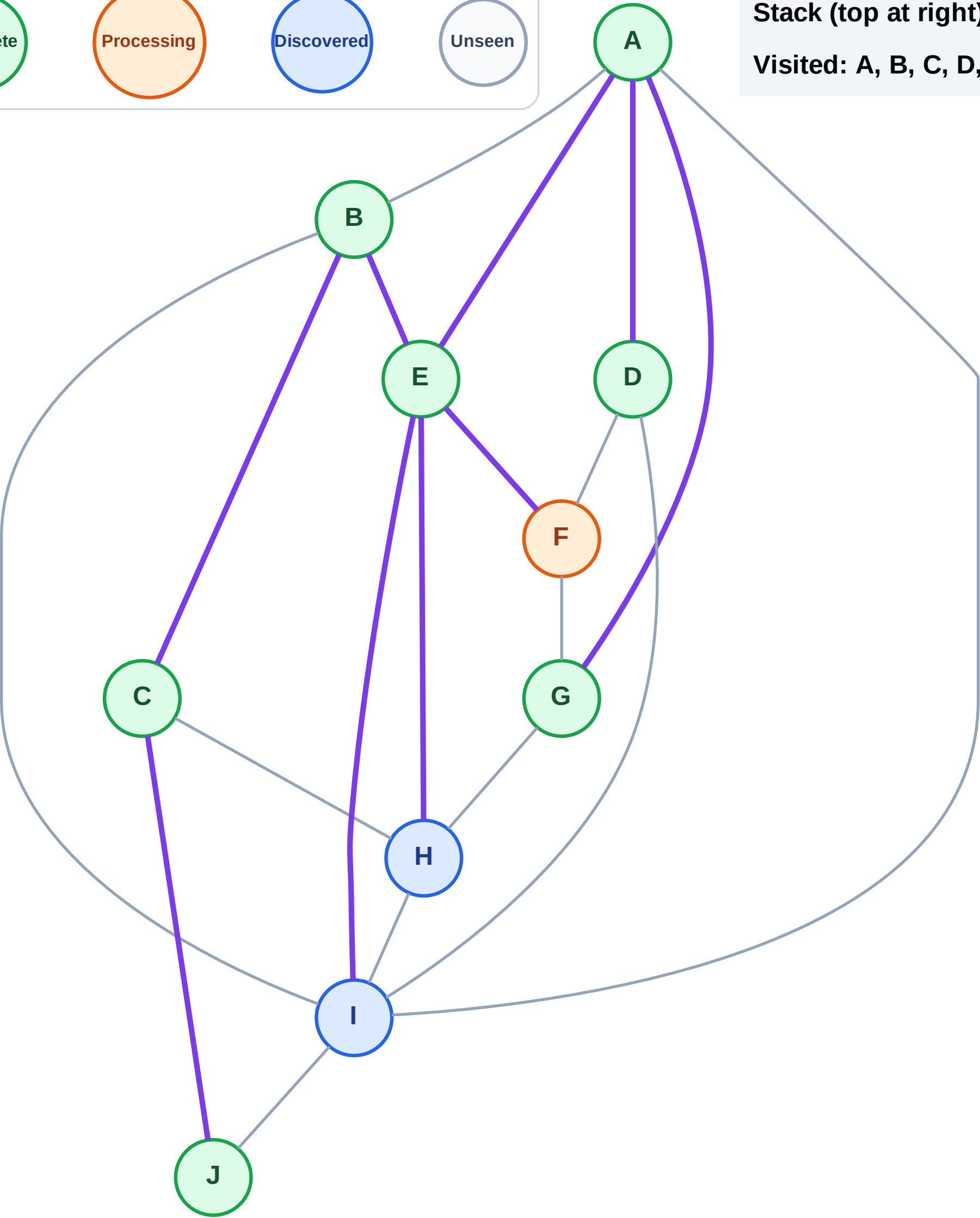
Complete

Processing

Discovered

Unseen

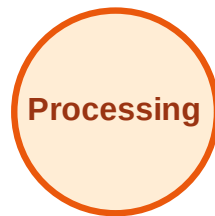
Stack (top at right): I | H
Visited: A, B, C, D, E, G, J



DFS Traversal

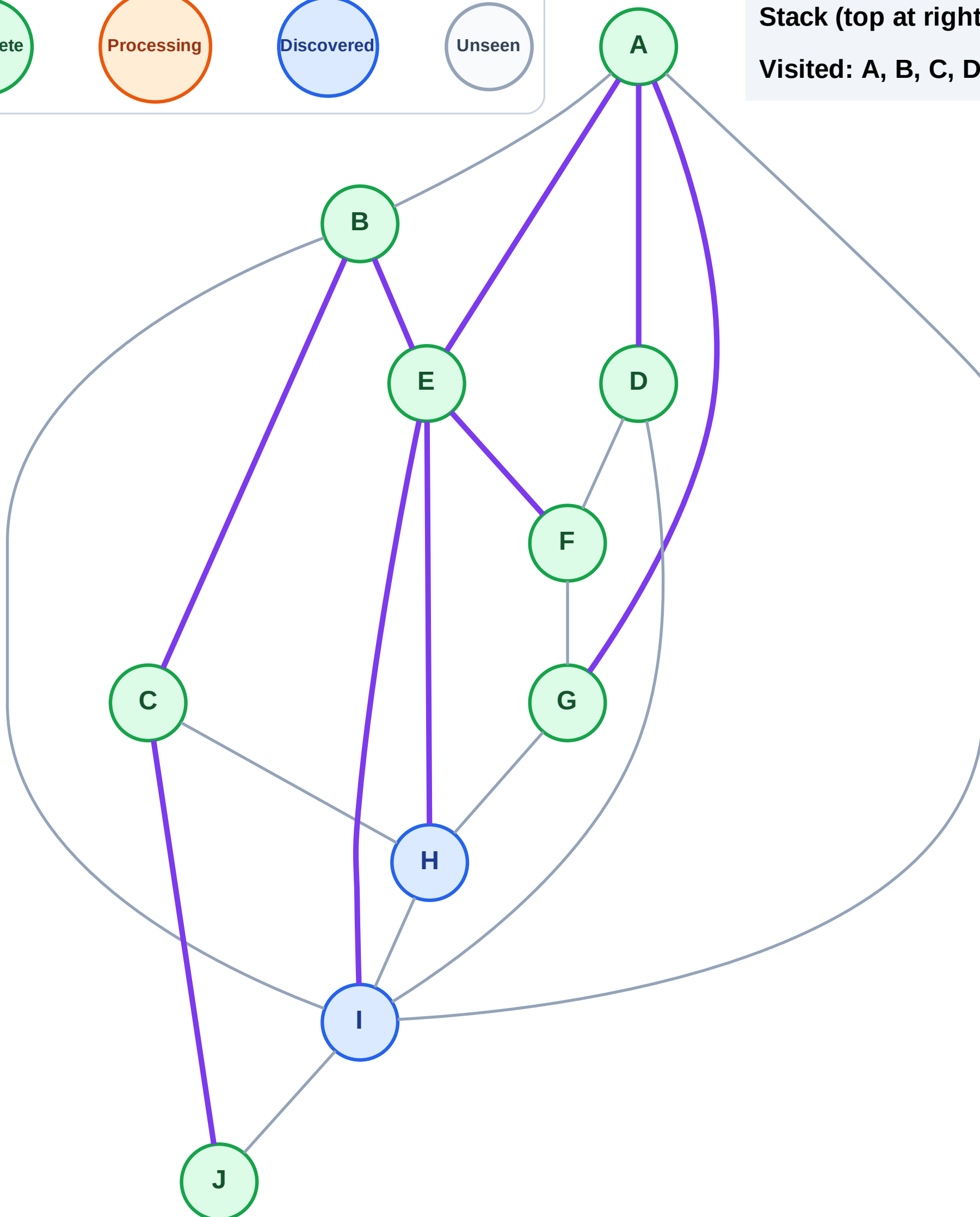
Step 17: No new neighbors from F

Legend



Stack (top at right): I | H

Visited: A, B, C, D, E, F, G, J



DFS Traversal

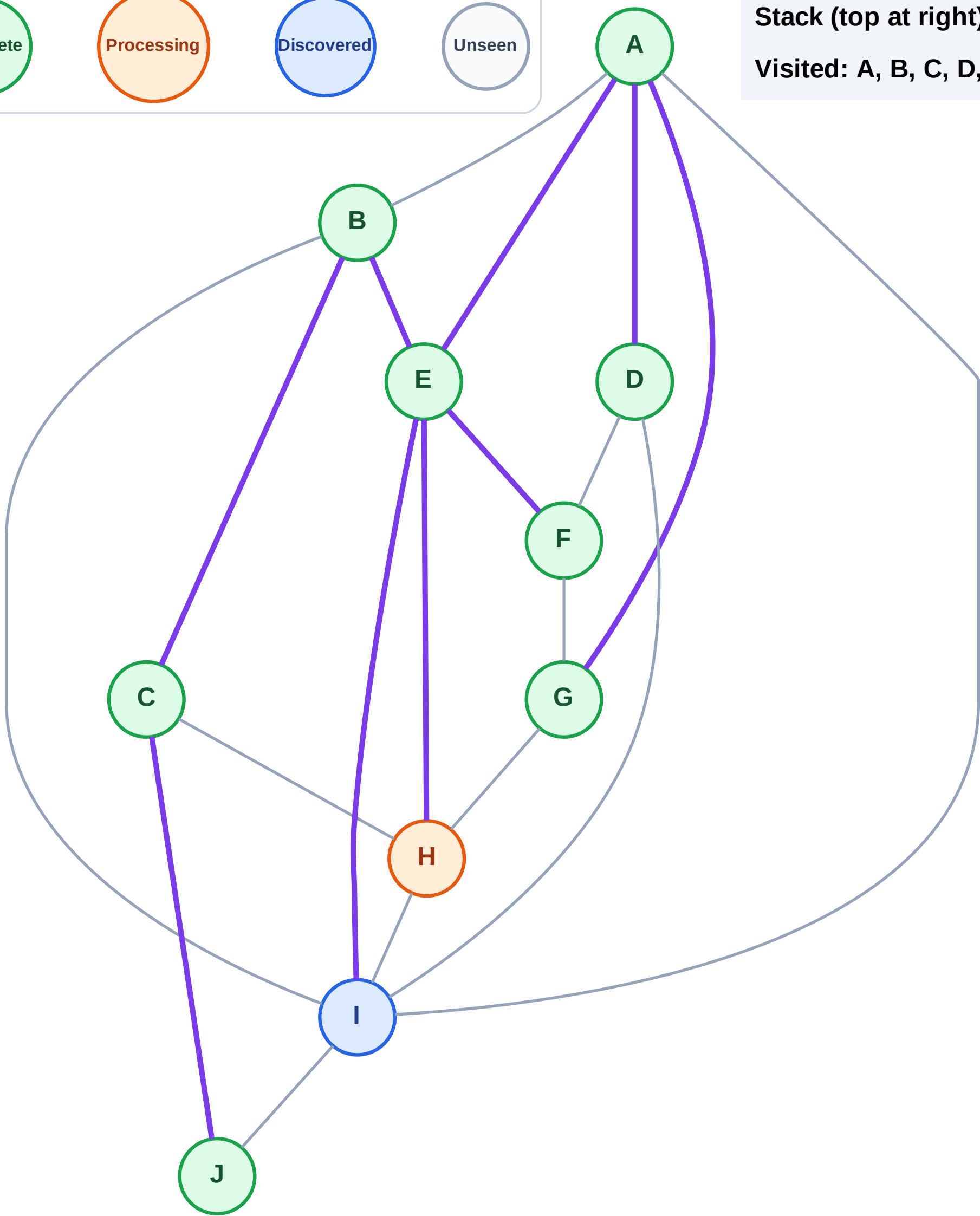
Step 18: Process node H

Legend



Stack (top at right): I

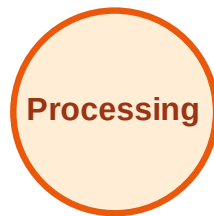
Visited: A, B, C, D, E, F, G, J



DFS Traversal

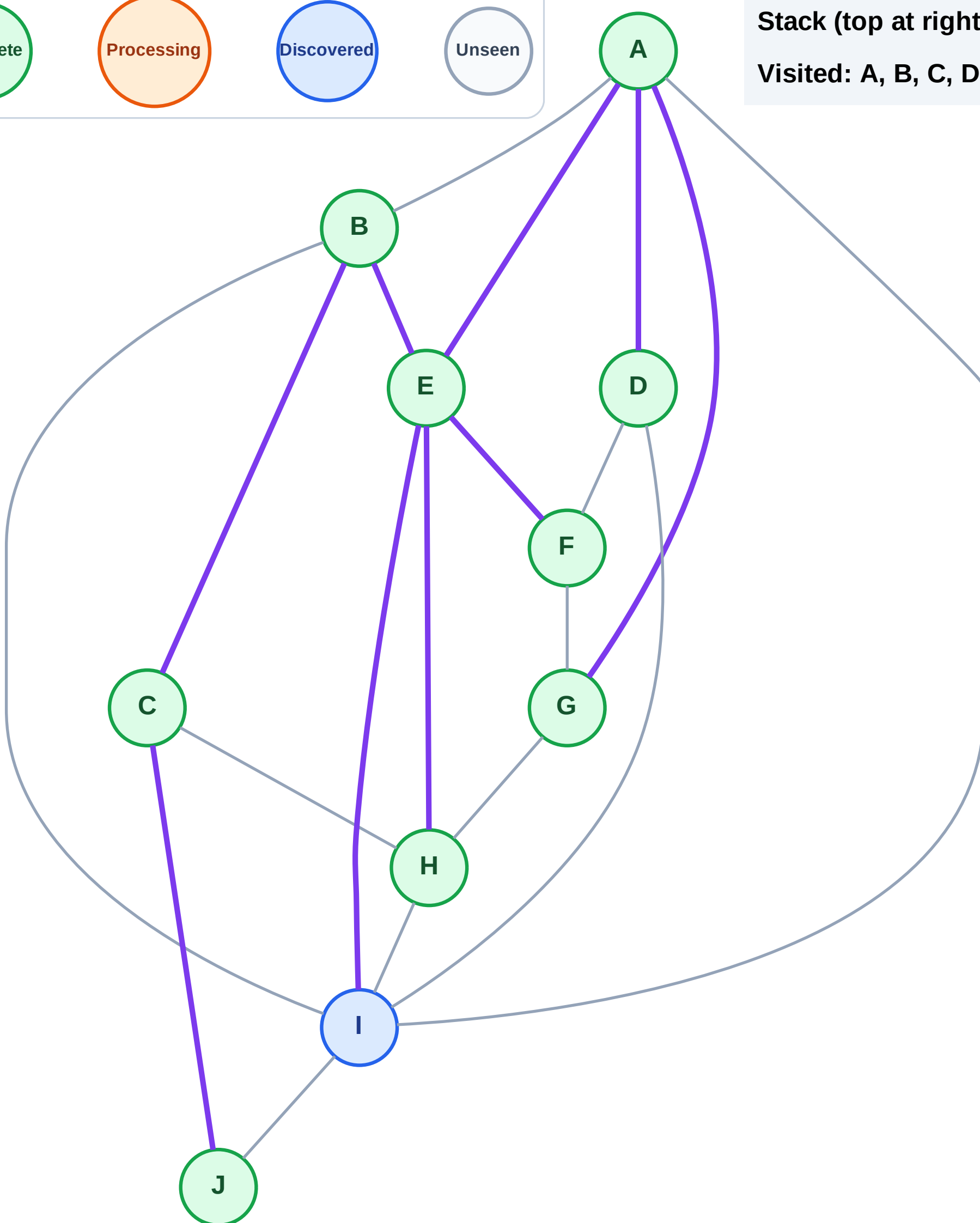
Step 19: No new neighbors from H

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

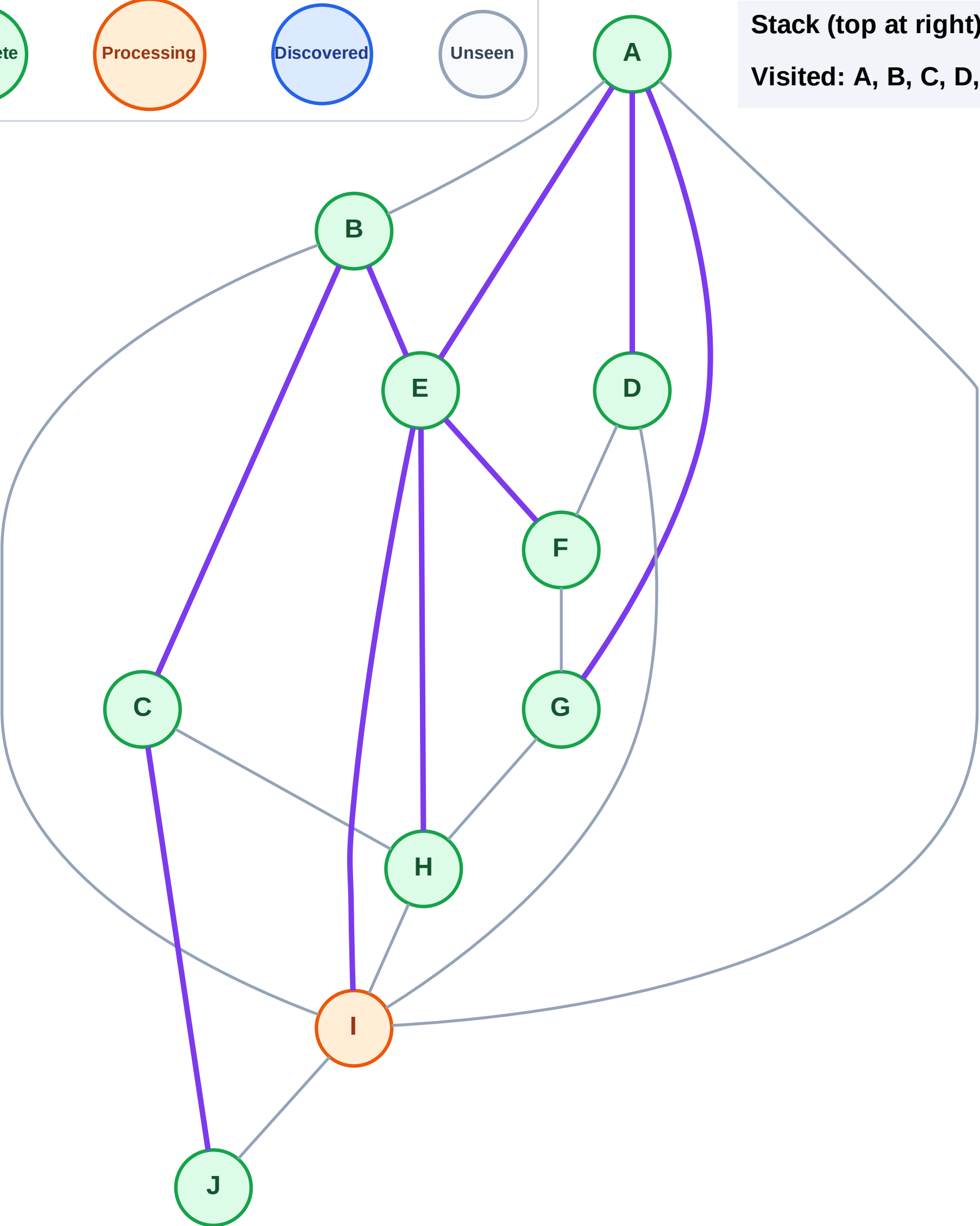
Step 20: Process node I

Legend



Stack (top at right): (empty)

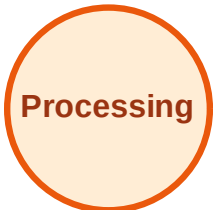
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

